Analysis pipeline

1. **What is data?** The most basic aspect of data is that it is a set. It could be for example a collection of patients, images, or sensor readings. The only information a set carries is its size. Data analysis is about discovering and studying various additional structures on the data set which can lead to understanding mechanisms behind the formation of the data and these structures. Therefore defining and studying various structures on sets is a vital part of this course. The focus in this course is on structures that reflect geometrical properties of data. The structures we are going to discuss are: partitions, pseudometrics, and simplicial complexes.

2. **Input.** In this course we concentrate on analysing data given in the following form:

$$\mathcal{I} = \{X_d \mid d \in \mathcal{D}\}$$

where

- $\mathcal{D}$ is the set we would like to analyse;
- for d in $\mathcal{D}$, X_d is a finite set representing the data point d.

We often assume the sets X_d, representing the data points in $\mathcal{D}$, are not "too small" and what that means will be illustrated later in the course with several examples. To discover various structures on the set $\mathcal{D}$, we are going to endow the sets X_d with structures that help in organizing information about $\mathcal{D}$. That is the main reason why X_d should not be too small.

In practice, X_d is often a finite subset of $\mathbf{R}^{n_d}$ where $\mathbf{R}$ denotes the set of real numbers and $\mathbf{R}^{n_d}$ the set of n_d-tuples of real numbers. Note that since we allow the dimensions n_d to vary, the vectors representing two distinct data points can have different length.

3. **Pipeline.** Our focus in this course is on the type of analysis whose objective is to discover a partition of D into clusters such as for example cases and controls. Our strategy is as follows:

- Endow the representing sets X_d with geometric structures such as pseudometrics or simplicial complexes.
- Define and study computable stable features, also called invariants, of these structures. We are interested in invariants for which we have means of comparing their values, and hence means of estimating how different the structures on X_d are for different data points d in D. In this course we concentrate on invariants constructed using homology. In fact one can think about this course as studying properties of homology.

Partitions

4. Let X be a finite set. A **partition** of X is a collection P of non-empty subsets of X such that:

- $\cup_{p \in P}\ p = X$ (they cover X)
- $p_1 \cap p_2 = \emptyset$ for $p_1 \neq p_2$ in P (they are disjoint).

A set X with a chosen partition P is denoted either as a pair (X, P), or as X if the partition is understood from the context, or as P if the set is understood from the context.

An element p in P is called a **block** or a **cluster** of the partition, the cardinality $|P|$ is called the **size** of the partition and the sequence of natural numbers $\{|p| \mid p \in P\}$, ordered in non-decreasing order, is called the **profile** of the partition. The profile is denoted either by $\overline{(X, P)}$ or by $\overline{X}$ or by $\overline{P}$. Note that the profile is a sequence of $|P|$ natural numbers. The sum of all the elements in the sequence $\overline{(X, P)}$ equals the number of elements $|X|$ in X.

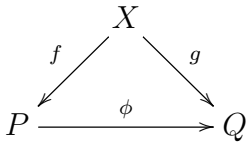
For example the partition $\{\{2, 4\}, \{1, 5\}, \{3\}\}$ of the set $\{1, 2, 3, 4, 5\}$ has size 3 and profile $1, 2, 2$. This set has $1 + 2 + 2 = 5$ elements.

The partition $S := \{\{x\} \mid x \in X\}$ is called the **singleton** partition of X and $T := \{X\}$ the **trivial** partition of X. The size of S equals the size of X and its profile is given by the sequence $1, 1, \ldots, 1$ of $|X|$ number of ones. For the trivial partition T, its size is 1 and its profile consists of a single natural number $|X|$.

The symbol $\mathrm{Part}(X)$ denotes the set of all partitions of X.

5. **Partitions and equivalence relations.** Let P be a partition of X. This partition induces a relation on the set X: two elements are said to be equivalent if they belong to the same cluster. This is an equivalence relation. Vice-versa, an equivalence relation on X gives a partition whose blocks are the equivalence classes of the relation. In this way partitions on X can be identified with equivalence relations on X. Via this identification, P corresponds to the set of the equivalence classes of the associated relation. The function $X \to P$, that maps an element x to the block $[x]$ containing it, is called the **quotient function**.

6. **Partitions and quotient functions.** Partitions can be in fact identified with such quotient functions. Let X be a set. We say that two surjective functions of sets $f\colon X \to P$ and $g\colon X \to Q$ are **identical** if there is a bijection $\phi\colon P \to Q$ for which the following triangle of functions commutes:



Note that if such a bijection ϕ exists, then it is unique.

Being identical is an equivalence relation on all surjections out of X. Equivalence classes of this relation are called **quotients** of X and we use the symbol $\mathrm{Quot}(X)$ to denote the set of all the quotients of X. By mapping a partition P of X to the quotient function $X \to P$ we obtain a function:

$$\mathrm{Part}(X) \to \mathrm{Quot}(X)$$

7. **Proposition.** *The function* $\mathrm{Part}(X) \to \mathrm{Quot}(X)$ *is a bijection.*

Proof. A quotient $f\colon X \to P$ leads to a partition of X given by the pre-images $\{f^{-1}(p) \mid p \in P\}$. This gives a function $\mathrm{Quot}(X) \to \mathrm{Part}(X)$ which is easily shown to be the inverse to $\mathrm{Part}(X) \to \mathrm{Quot}(X)$, proving the proposition. □

8. We identify the following structures on a set X:
   (1) partitions of X;
   (2) equivalence relations on X;
   (3) quotients of X.

9. **Comparing partitions of the same set.** How can we compare partitions of the same set? Let P and Q be two partitions of X. The partition P is called a **sub-partition** of Q or **refinement** of Q, denoted by $P \geq Q$, if, for any block p in P, there is a block q in Q, such that $p \subseteq q$. For example the singleton partition S of X is a sub-partition of any other partition of X. Analogously any partition of X is a sub-partition of the trivial partition T. Thus for any partition P of X we have $S \geq P \geq T$.

Note that if $P \geq Q$, then $|P| \geq |Q|$. Furthermore $P \geq Q$ and $Q \geq P$ happens if and only if $P = Q$. Moreover $P \geq Q \geq R$ implies $P \geq R$.

10. **Comparing partitions of different sets.** It is also important to be able to compare partitions of different sets. Let P be a partition of X, Q be a partition of Y. A function $f\colon X \to Y$ **preserves partitions** if, for any block p in P, there is a block q in Q such that $f(p) \subseteq q$. Any function $f\colon X \to Y$ which preserve partitions induces a function, denoted by the same symbol $f\colon P \to Q$, that maps p in P to the necessary unique q in Q for which $f(p) \subseteq q$. For example if P and Q are partitions of the same set X, then $\mathrm{id}\colon (X, P) \to (X, Q)$ preserves partitions if and only if $P \geq Q$.

11. **Exercise.** Let (X, P) and (Y, Q) be finite sets with partitions. Prove that the following are equivalent, in which case we say that (X, P) and (Y, Q) are **isometric**:
   (1) There is a bijection $f\colon X \to Y$ such that p belongs to P if and only if $f(p)$ belongs to Q.
   (2) The profiles $\overline{(X, P)}$ and $\overline{(Y, Q)}$ are equal

12. **Exercise.** Let (X, P) and (Y, Q) be finite sets with partitions. Prove that the following are equivalent:
   (1) There is a function $f\colon X \to Y$ preserving partitions for which the induced function $f\colon P \to Q$ is a bijection.

(2) $|P| = |Q|$.

Distances

13. The symbol $\mathbf{R}$ or $(-\infty, \infty)$ denotes the set of real numbers with the usual arithmetic operations and order relation. The symbol $[0, \infty)$ denotes the set of non-negative numbers and $(-\infty, \infty]$ and $[0, \infty]$ denote respectively the disjoint unions $\mathbf{R} \cup \{\infty\}$ and $[0, \infty) \cup \{\infty\}$. We extend the standard arithmetic operations and the order relation on $\mathbf{R}$ to $(-\infty, \infty]$ and $[0, \infty]$ as follows:

$$r + \infty = \infty \text{ and } r \leq \infty \text{ for all } r \text{ in } (-\infty, \infty]$$

$$\infty - \infty = 0 \text{ and } \infty - r = \infty \text{ for all } r \text{ in } (-\infty, \infty)$$

14. **Distances.** Let X be a set. A **distance** on X is a function:

$$d\colon X \times X \to [0, \infty]$$

satisfying the following conditions for x and y in X:

(1) symmetry: $d(x, y) = d(y, x)$;
(2) reflexivity: $d(x, x) = 0$;

A distance is called a **pseudometric** if in addition it satisfies:

(3) **triangle inequality**: $d(x, y) + d(y, z) \geq d(x, z)$ for x, y, and z in X.

A pseudo-metric is called a **metric** if in addition it satisfies:

(2)' $d(x, y) = 0$ if and only if $x = y$.

A set with a chosen distance is called a **distance space** and is denoted either as a pair (X, d), or as X if the distance is understood from the context, or as d if the set is understood from the context. If the distance d is a (pseudo)metric, then (X, d) is called a **(pseudo)metric space**. If the set X is finite, then (X, d) is a called a **finite distance/(pseudo)metric space**. The symbols $\mathcal{D}(X)$, $\mathcal{PM}(X)$ and $\mathcal{M}(X)$ denote respectively the sets of all distances, pseudometrics and metrics on X. There are inclusions:

$$\mathcal{M}(X) \subset \mathcal{PM}(X) \subset \mathcal{D}(X)$$

15. **Restrictions.** Let (X, d_X) be a distance space and $f\colon Y \to X$ b e a function. Define $d_Y(y_1, y_2) := d_X(f(y_1), f(y_2))$ for y_1 and y_2 in Y. Then (Y, d_Y) is a distance space. It is called the **restriction** of d_X along f.

If d_X is a pseudometric, then so is any of its restrictions. Is the same true for a metric?

Let $Y \subset X$ be a subset. The restriction of d_X along such an inclusion is called a subspace of (X, d_X). Note that any subspace of a metric space is also a metric space.

Finite pseudometric spaces are the most important object of study in this course. As we will see, the triangle inequality condition required for pseudometric spaces is important for assuring the stability of the invariants we are going to construct.

16. **Exercise.** One might ask why to measure distances we do often insist on assuming triangular inequality? We hope that an answer to this question will become apparent by the end of this course. See Exercise 47 for an important example of a distance and not a pseudometric. Give 3 reasons why you think triangular inequality is needed or not needed.

17. **Separation and diameter.** Let X be a finite set. If the distance d on X has a positive real value, then the minimal positive real value of d is called the **separation** of d. In case d has no positive real values (only 0 and ∞ occur as its values), we declare its separation to be 0. The separation is denoted by $\mathrm{sep}(d)$ or simply by sep if the distance d is assumed. The maximal value of d is called the **diameter** of d and is denoted by $\mathrm{diam}(d)$ or simply by diam if the distance d is assumed.

18. **Balls and disks.** Let d be a distance on X and x be en element in X. The following subsets of X are called respectively a **ball** and a **disk** of radius r around x:

$$B(x,r) := \{y \in X \mid d(x,y) < r\} \qquad D(x,r) := \{y \in X \mid d(x,y) \leq r\}$$

For a subset $Z \subset X$, **open and closed** r **neighbourhoods** are defined as follows:

$$B(Z,r) := \cup_{z\in Z} B(z,r) = \{y \in X \mid d(y,z) < r \text{ for some } z \text{ in } Z\}$$

$$D(Z,r) := \cup_{z\in Z} D(z,r) = \{y \in X \mid d(y,z) \leq r \text{ for some } z \text{ in } Z\}$$

19. **Distance matrices.** One way to encode a distance on a finite X is as follows:

- Enumerate elements of X to get $X = \{x_1, \ldots, x_k\}$.
- Encode distances by a $k \times k$ matrix $D := (d(x_i, x_j))_{1\leq i,j\leq k}$.

This matrix D, called the **distance matrix** of the finite distance space X, is symmetric with values in $[0,\infty]$ and with 0's on the diagonal. These are the consequences of conditions (1) and (2) in Point 14. Any such matrix (symmetric with 0 diagonal) is a distance matrix of some distance. By enumerating elements of X, we can therefore identify distances on X with $k \times k$ matrices with values in $[0,\infty]$ which are symmetric and have 0 diagonal.

Assume $k \geq 2$ (there is at least two distinct points in X). Since the distance matrix is symmetric, it is more efficient to store it as a vector of length $\binom{k}{2}$:

$$D \mapsto (D_{21}, D_{31}, D_{32}, D_{41}, D_{42}, D_{43}, \ldots, D_{k1}, \ldots, D_{k,k-1})$$

In this transformation $D_{i,j} = d(x_i, x_j)$ is placed on the $\frac{(i-1)(i-2)}{2} + j$ coordinate for any $1 \leq j < i \leq k$.

20. **Exercise.** Describe the reverse procedure, to the above transformation, from a vector into the corresponding distance matrix.

21. **From distances to pseudometrics.** Let d be a distance on X. For x and y in X define:

$$Ld(x,y) := \inf\{d(x_0,x_1) + \cdots + d(x_{k-1},x_k) \mid \text{for all sequences } x = x_0, x_1, \ldots, x_k = y\}$$

Then Ld is a pseudometric on X. By assigning to a distance d on X the pseudometric Ld on X we obtain a function which is called **linkage**:

$$L\colon \mathcal{D}(X) \to \mathcal{PM}(X)$$

22**. Exercise.** Show that Ld is indeed a pseudometric on X. Show further, that if d is a pseudometric, then $d = Ld$. Conclude that we have the following commutative diagram:

$$\mathcal{PM}(X) \hookrightarrow \mathcal{D}(X) \xrightarrow{L} \mathcal{PM}(X), \qquad \mathcal{PM}(X) \xrightarrow{\text{id}} \mathcal{PM}(X)$$

23**. Exercise.** Let d be a distance on X. Show that Ld is a metric on X if and only if d satisfies the following condition: $d(x, y) = 0$ if and only if $x = y$.

Discrete distances and pseudometrics

24. We often think about a distance on X as encoding a relation of being close and being apart for points in X. A distance gives numerical values (obtained by taking measurements of outcomes of some experiments or observations) to such relations. Although convenient for encoding, interpreting and making adequate decisions based on arbitrary distances is often difficult and not straightforward. This course is in fact about interpreting distance relations and describing possible outcomes of such interpretations into mathematical structures which are easier to comprehend, visualise and hence simpler to interpret. The most basic and often the most desirable structure we look for is a partition of X. Based on measurements, we would like to partition X into groups that reflect some ground truth about the experiment. Partitions are easily interpretable and visualisable. In this section we explain how to use a distance to produce various partitions of X.

25. **From distances to partitions.** Let d be a distance on X. Choose t in $[0, \infty)$. Two points x and y in X are said to be **directly t equivalent** with respect to d if $d(x, y) \leq t$. Being directly 0 equivalent is described simply as being **directly equivalent**.

Being directly t equivalent is a symmetric and reflexive relation on X, however it may fail to be transitive. The equivalence relation generated by the relation of being directly t equivalent is called being t **equivalent** with respect to d. We also use the term **indistinguishable** with respect to d for being 0 equivalent. Explicitly, two elements x and y in X are t equivalent with respect to d if there is a sequence $x = x_0, x_1, \ldots, x_k = y$ of elements in X such that $d(x_i, x_{i+1}) \leq t$ for $i = 0, \ldots, k-1$. This is an equivalence relation. We denote by $[x]_t \subset X$ the subset of all elements in X which are t equivalent to x, i.e., the equivalence class that contains x. These subsets form a partition of X and we denote by X/d_t the set of equivalence classes of this relation. Thus any t leads to a function $\mathcal{D}(X) \to \mathrm{Part}(X)$ that maps a distance d on X to the partition $(X, X/d_t)$. The partition X/d_0 is also denoted simply by X/d and we use the symbol $\pi_0 \colon \mathcal{D}(X) \to \mathrm{Part}(X)$ to denote the function that maps a distance d on X to the partition $(X, X/d)$.

26. **Exercise.** Let d be a distance on X. Recall that we can form a pseudometric Ld on X (see point 21). Show that, for any $t \geq 0$, the partitions X/d_t and X/Ld_t are equal. Conclude that the function π_0 fits into the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{D}(X) & \xrightarrow{L} & \mathcal{PM}(X) & \xrightarrow{\pi_0} & \mathrm{Part}(X) \\ & \searrow^{\pi_0} & & \nearrow & \end{array}$$

27. **Exercise.** Show that if d is a pseudometric on X, then being directly equivalent is an equivalence relation on X (thus for a pseudometric, being directly equivalent is

the same as being 0 equivalent or indistinguishable). Give an example of a distance for which being directly equivalent is not an equivalence relation. Give an example of a pseudometric d on X and $t > 0$ for which being directly t equivalent is not an equivalence relation.

28. **Exercise.** Let d be a distance on X. Show that the quotient function $X \to X/d$ is a bijection if and only if Ld is a metric.

29. **Discrete distances.** A distance d on X is called **discrete** if, for x and y in X, either $d(x, y) = 0$ or $d(x, y) = \infty$. The symbols $\mathrm{Dis}\mathcal{D}(X)$ and $\mathrm{Dis}\mathcal{PM}(X)$ denote respectively the set of discrete distance and discrete pseudometrics on X. Note that if d is a discrete distance on X then Ld (see Point 21) is a discrete pseudometric on X.

Let t be in $[0, \infty)$. For an arbitrary distance d on X, define a discrete distance d_t on X as follows:

$$d_t(x, y) := \begin{cases} 0 & \text{if } d(x, y) \leq t \\ \infty & \text{otherwise} \end{cases}$$

30. **Exercise.** Show that a distance d on X is discrete if and only if its separation (see Point 17) is 0.

31. **Exercise.** Show that if d is a pseudometric, then so is d_0. Give an an example of a pseudometric d and $t > 0$ for which d_t is not a pseudometric.

32. **Exercise.** Let d be a discrete distance on X. Prove that the following statements are equivalent:

- being directly equivalent with respect to d is an equivalence relation on X;
- d is a pseudometric.

33. **Exercise.**

(1) Show that if d is a discrete distance on X, then being directly 0 equivalent is the same as being directly t equivalent, for all t in $[0, \infty)$.
(2) A discrete distance leads to a symmetric and reflective relation of being directly 0 equivalent on X. Show that any symmetric and reflexive relation can be obtained in this way. Conclude that there is a bijection between the set of discrete distances on X and the set of symmetric and reflective relations on X.
(3) Show that via the above identification of discrete distances with symmetric and reflective relations discrete pseudometric correspond to equivalence relations.

Based on this exercise, we identify discrete distances on X with symmetric and reflective relations on X and discrete pseudometric on X with equivalence relations on X.

(4) Show that via this identification, the formation of the equivalence relation generated by a symmetric and reflective relation corresponds to the construction L of a pseudometric out of an arbitrary distance (see Point 21).

34. **Exercise.** Let d be a distance on X and t be in $[0, \infty)$. According to our notation the symbol X/d_t can be interpreted in two ways:

- X/d_t is the partition of X given by the equivalence relation of being t-equivalent with respect to d.
- X/d_t is the partition of X given by the equivalence relation of being 0-equivalent with respect to d_t.

Prove that these two partitions of X are equal and hence these two interpretations coincide.

35. **Proposition.** *The following composition is a bijection:*

$$\mathrm{Dis}\mathcal{PM}(X) \subset \mathcal{PM}(X) \xrightarrow{\pi_0} \mathrm{Part}(X)$$

Proof. Let d and d' be discrete pseudometrics. Note that $\pi_0(d) = \pi_0(d')$ is equivalent to $d(x, y) = 0$ if and only if $d'(x, y) = 0$. Since d and d' are discrete, this means $\pi_0(d) = \pi_0(d')$ if and only if $d = d'$. The composition in the statement of the proposition is therefore an injection.

Let P be a partition of X. Define:

$$d(x, y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ belong to the same block in } P \\ \infty & \text{otherwise} \end{cases}$$

Then d is a discrete pseudometric for which $X/d = P$. The composition is therefore also surjective. □

36. **Exercise.** Show that the distance defined in the proof of Proposition 35 is indeed a pseudometric.

37. **Partitions and discrete pseudometrics.** We use Proposition 35 to identify partitions with discrete pseudometrics. Via this identification the function $\pi_0 \colon \mathcal{D}(X) \to \mathrm{Part}(X) = \mathrm{Dis}\mathcal{PM}(X)$ maps a distance d to a discrete pseudometric given by Ld_0.

38. Recall (see Point 8) that partitions of X are identified with equivalence relations and quotients of X. Above we have seen how partitions of X can also be identified with discrete pseudometrics on X. From now on we are not going to distinguish between these structures on X: partitions, equivalence relations, quotients and discrete pseudometrics. Since being able to translate between these structures is important we summarise here how it is done explicitly.

(1) Let P be a partition of X.
   - P corresponds to the equivalence relation on X where x and y are related if and only if they belong to the same block of P.
   - The function $X \to P$ that maps x to its block is the associated quotient function.

- The discrete pseudometric associated to P is given by:

$$d(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ belong to the same block} \\ \infty & \text{otherwise} \end{cases}$$

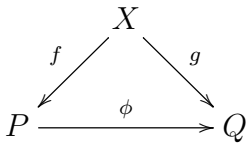
(2) Let d be a discrete pseudometric on X.
- The associated partition X/d is given by subsets $[x] = \{y \mid d(x,y) = 0\}$.
- The function $X \to X/d$ that maps x to the block $[x]$ is the quotient function corresponding to d.
- The associated equivalence relation relates x and y if $d(x,y) = 0$.

(3) Let $f\colon X \to P$ be a quotient function.
- The associated partition is given by the preimages $f^{-1}(p)$ for p in P.
- The corresponding equivalence relation relates x and y if and only if $f(x) = f(y)$.
- The discrete pseudometric associated to f is given by:

$$d(x,y) = \begin{cases} 0 & \text{if } f(x) = f(y) \\ \infty & \text{otherwise} \end{cases}$$

39. **Exercise.** Recall that a partition P of X is a sub partition of Q (denoted by $P \geq Q$) if any block of P is included in some block of Q. The corresponding notions for discrete pseudometrics and quotients are called **sub metrics** and **sub quotients**. If a discrete pseudometric d is a sub metric of another discrete pseudometric d' we denote this relation by $d \geq d'$. We use the same symbol to denote sub quotients.

- Let d and d' be discrete pseudometrics on X. Show that d is a sub metric of d' if and only if the following implication holds: $d(x,y) = 0 \Rightarrow d'(x,y) = 0$.
- Let $f\colon X \to P$ and $g\colon X \to Q$ be quotient functions. Prove that f is a sub quotient of g if and only if there is a function $\phi\colon P \to Q$ for which the following diagram commutes (note that such a function is necessarily surjective):



Decomposing a pseudometric into a metric and a partition

40. **Metric induced by a pseudometric.** Let d be a pseudometric on X. According to Exercise 27, being directly equivalent with respect to d is therefore the same as being equivalent. Hence the partition X/d consists of subsets of the form $[x]$ consisting of elements z in X such that $d(x, z) = 0$.

We claim that if $d(x, x') = 0$ and $d(y, y') = 0$, then $d(x, y) = d(x', y')$. This follows from the triangle inequality:

$$d(x, y) \leq d(x, x') + d(x', y) \leq d(x, x') + d(x', y') + d(y', y) = d(x', y')$$

Consequently the function $d\colon X/d \times X/d \to [0, \infty]$, that maps $([x], [y])$ to $d(x, y)$, is well defined as it does not depend on the choice of x and y in the blocks $[x]$ and $[y]$.

41. **Proposition.** *Let d be a pseudometric on X. Then the function $d\colon X/d \times X/d \to [0, \infty]$ is a metric.*

Proof. Since $d\colon X \times X \to [0, \infty]$ is a pseudometric, then so is $d\colon X/d \times X/d \to [0, \infty]$. If $d([x], [y]) = 0$, then $d(x, y) = 0$ and hence $[x] = [y]$ which shows that $d\colon X/d \times X/d \to [0, \infty]$ is in fact a metric. □

With a pseudometric $d\colon X \times X \to [0, \infty]$ we have associated two objects: a partition X/d of the set X and a metric $d\colon X/d \times X/d \to [0, \infty]$ on this partition. These two objects determine the pseudometric d:

42. **Theorem.** *Let X be a set.*

(1) *Let d and d' be pseudometrics on X. Then $d = d'$ if and only if the following conditions are satisfied:*
   - *the partitions X/d and X/d' of X are the same.*
   - *the metrics $d\colon X/d \times X/d \to [0, \infty]$ and $d'\colon X/d' \times X/d' \to [0, \infty]$ are the same.*

(2) *Furthermore, for any partition P of X and any metric d' on P, there is a unique pseudometric d on X such that $X/d = P$ and $d = d'$ on P.*

43. **Exercise.** Prove Theorem 42.

Metrics on $\mathbf{R}^n$

It is in general not a good idea to define a pseudometric on X by explicitly specifying a distance matrix and then verifying that triangle inequality holds. Assuring triangle inequality is in fact often expensive. A better strategy is to map X into a known (pseudo)metric space and then consider the restriction of the (pseudo)metric to X. For this we need a zoo of examples of (pseudo)metric spaces which we can use as targets. In this section we describe the most popular **target metric spaces** which are metrics on the set of n-tuples of real numbers $\mathbf{R}^n$.

44. The symbol $\mathbf{R}^n$ denotes the set of n-tuples of real numbers. For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbf{R}^n$:

- The number $x_1y_1 + \cdots x_ny_n$ is denoted by $x \cdot y$ or simply by xy and called the **scalar product**.
- The number $\sqrt{x_1^2 + \cdots + x_n^2}$ is denoted by $\|x\|$ and called the **norm** of x.
- Let $p > 0$. The number $(|x_1|^p + \cdots + |x_n|^p)^{1/p}$ is denoted by $\|x\|_p$ and called the p**-norm** of x.
- The number $|\{i \mid x_i \neq 0\}|$ is denoted by $\|x\|_0$ and called the 0**-norm** of x.
- The number $\max\{|x_1| \ \ldots, |x_n|\}$ is denoted by $\|x\|_\infty$ and called the ∞**-norm** of x.

45. The following are examples of metrics on the set $\mathbf{R}^n$:

- **Euclidean metric, called also the l_2 metric:**
$$d_2(x, y) = \|x - y\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$
- **Cityblock metric, called also the l_1 metric:**
$$d_1(x, y) = \|x - y\|_1 = |x_1 - y_1| + \cdots + |x_n - y_n|$$
- **Hamming metric, called also the l_0 metric:**
$$d_0(x, y) = \|x - y\|_0 = |\{i \mid x_i \neq y_i\}|$$
- l_p **metric for $p > 0$:**
$$d_p(x, y) = \|x - y\|_p = (|x_1 - y_1|^p + \cdots + |x_n - y_n|^p)^{1/p}$$
- **Chebyshev metric:**
$$d_\infty(x, y) = \|x - y\|_\infty = \max\{|x_1 - y_1|, \ldots, [x_n - y_n|\}$$
It turns out that:
$$d_\infty(x, y) = \lim_{p\to\infty} [(x_1 - y_1)^p + \cdots + (x_n - y_n)^p]^{1/p}$$

46. The following are examples of distances on the set $\mathbf{R}^n$:

- **Cosine distance:**

$$d(x,y) = \begin{cases} 1 - \frac{x_1y_1+\cdots x_ny_n}{\sqrt{x_1^2+\cdots x_n^2}\sqrt{y_1^2+\cdots y_n^2}} = 1 - \frac{x\cdot y}{\|x\|\|y\|} & \text{if } x \neq 0 \text{ and } y \neq 0 \\ \infty & \text{otherwise} \end{cases}$$

- **Correlation distance:**

$$d(x,y) = 1 - \frac{x_c \cdot y_c}{\|x_c\|\|y_c\|}$$

where $x_c = (x_1 - \frac{x_1+\cdots x_n}{n}, \ldots, x_n - \frac{x_1+\cdots x_n}{n})$ is called x **centered**.

47**. Exercise.** Show that neither cosine nor correlation distance satisfies the triangular inequality. Conclude that these distances are not pseudometrics $\mathbf{R}^n$:

One might ask why to measure distances we often insist on assuming triangular inequality? We hope that an answer to this question will become apparent by the end of this course.

48**. Proposition.** *For $\infty \geq q > p > 0$ and x and y in $\mathbf{R}^n$:*

$$d_q(x,y) \leq d_p(x,y)$$

Proof. We are going to use the following lemma:

Assume that $f\colon \mathbf{R} \to \mathbf{R}$ is such that $f(x)/x$ is a non-decreasing function for $x > 0$. Then $f(x_1 + \cdots + x_n) \geq f(x_1) + \cdots + f(x_n)$ for any $x_i > 0$.

If we set $g(x) = f(x)/x$, then $f(x) = xg(x)$. Since $g(x)$ is a non-decreasing function for $x > 0$ we have the following inequalities for any $x_i > 0$:

$$f(x_1 + \cdots + x_n) = (x_1 + \cdots + x_n)g(x_1 + \cdots + x_n) =$$
$$= x_1g(x_1 + \cdots + x_n) + \cdots + x_ng(x_1 + \cdots + x_n) \geq x_1g(x_1) + \cdots + x_ng(x_n) =$$
$$= f(x_1) + \cdots + f(x_n)$$

To prove the proposition it is enough to show that for $x_i > 0$, if $q > p > 0$, then

$$(x_1^q + \cdots + x_n^q)^{1/q} \leq (x_1^p + \cdots + x_n^p)^{1/p}$$

This inequality is equivalent to:

$$x_1^q + \cdots + x_n^q \leq (x_1^p + \cdots + x_n^p)^{q/p}$$

Since for $r > 1$, the function $x \mapsto x^r/x = x^{r-1}$ is non-decreasing, the last inequality is a direct consequence of the lemma. □

49**.** Note that the Euclidean, cityblock, hamming, l_p, and Chebyshev metrics can be extended to the set of n-tuples of extended set of real numbers $(-\infty, \infty]^n$.

Metrics on sets

Another popular target metric spaces are the spaces of all subsets of a given set with symmetric difference and Jaccard metrics.

50. Consider the set $P(X)$ of all finite subsets of a given set X. For A and B in $P(X)$, define:

- **Symmetric difference metric:**
$$d(A,B) = |A \Delta B| = |(A \setminus B) \cup (B \setminus A)|$$
- **Jaccard metric.**
$$d(A,B) = \begin{cases} 0 & \text{if } A = B = \emptyset \\ 1 - \frac{|A \cap B|}{|A \cup B|} = \frac{|A \Delta B|}{|A \cup B|} = \frac{|(A \setminus B) \cup (B \setminus A)|}{|A \cup B|} & \text{otherwise} \end{cases}$$

51. **Exercise.** Prove that the symmetric difference metric is a metric on $P(X)$.

To prove that Jaccard metric is a metric one uses a so called Steinhaus Transform:

52. **Proposition.** *Let d be a (pseudo)metric on a set X. Choose an element x in X. Then the following function, called* **the Steinhaus Transform***, is also a (pseudo) metric:*
$$d'(a,b) = \begin{cases} \infty & \text{if } d(a,b) = \infty \\ 0 & \text{if } d(a,x) = d(b,x) = 0 \\ \frac{2d(a,b)}{d(a,x)+d(b,x)+d(a,b)} & \text{otherwise} \end{cases}$$

53. **Exercise.** Prove Proposition 52.

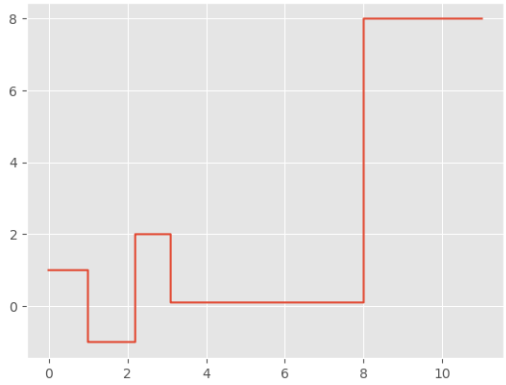
54. **Exercise.** Prove that the Jaccard metric is the Steinhaus Transform of the symmetric difference metric when the empty subset is chosen. Conclude that the Jaccard metric is in fact a metric.

Piece-wise constant functions

In this section we describe a target metric space that we are going to use for homological invariants.

55. A function $f\colon [0,\infty) \to (-\infty,\infty)$ is called **piece-wise constant** (a pcf for short) if there is a finite number of elements $0 = a_0 < a_1 < \cdots < a_n$ in $[0,\infty)$ for which the restrictions of f to the right open intervals $[a_0,a_1),\ldots,[a_{n-1},a_n),[a_n,\infty)$ are constant functions. We also say that the sequence $0 = a_0 < a_1 < \cdots < a_n$ **discretises** f. The value $f(a_n)$ is denoted by $\lim(f)$. Note that the sequence that discretises a pcf is not unique. For example if $0 < 1$ discretises a given pcf then so does $0 < 0.5 < 1 < 2$.

Here is an example of the graph of a pcf which is discretised by a sequence $0 < 1 < 2.2 < 3.1 < 8$. This pcf is also discretised by $0 < 0.5 < 1 < 2 < 2.2 < 3.1 < 8$.



Note that if f and g are pcfs and r is a real number, then the functions $f+g$, $f-g$, rf, and $|f|$ are also pcfs. In particular the set of pcfs form a vector space over the field of real numbers $\mathbf{R}$.

56. **Integrals.** Let f be a pcf. We denote by $\int_0^\infty f$ the Riemann integral of f over the interval $[0,\infty)$. For a pcf f this integral can be calculated as follows. Choose a sequence $0 = a_0 < a_1 < \cdots < a_n$ that discretises f. Then:

$$\int_0^\infty f = \begin{cases} \infty & \text{if } \lim(f) > 0 \\ -\infty & \text{if } \lim(f) < 0 \\ (a_1 - a_0)f(a_0) + \cdots + (a_n - a_{n-1})f(a_{n-1}) & \text{if } \lim(f) = 0 \end{cases}$$

57. **Metrics on PCF's.** The following are examples of metrics on the set of pcf's:

- l_p **metric for** $\infty > p \geq 1$**:**

$$d_p(f,g) = \left(\int_0^\infty |f-g|^p\right)^{1/p}$$

- **Interleaving metric:**

$$d_{\bowtie}(f,g) = \begin{cases} \infty & \text{if } \{v \mid f(x) \geq g(x+v) \text{ and } g(x) \geq f(x+v) \text{ for any } x\} = \emptyset \\ \inf\{v \mid f(x) \geq g(x+v) \text{ and } g(x) \geq f(x+v) \text{ for any } x\} & \text{otherwise} \end{cases}$$

We say that f and g are ϵ **interleaved** if $d_{\bowtie}(f,g) \leq \epsilon$.

58. **Exercise.** Prove that indeed d_p and $d_{\bowtie}$ are metrics between pcfs.

59. **Proposition.** *For $q > p \geq 1$ and pcfs f and g:*

$$d_q(f,g) \leq d_p(f,g)$$

Proof. This is a direct consequence of Proposition 48. □

Let $0 < 1 < 2$ be a sequence that discretises both f and g. Assume $f(0) = 0$, $f(1) = 1$, $f(2) = 2$, $g(0) = 2$, $g(1) = 1$ and $g(2) = 2$. Since $f(0) < g(x)$ for any x, the following set is empty:

$$\{v \mid f(x) \geq g(x+v) \text{ and } g(x) \geq f(x+v) \text{ for any } x\}$$

Thus $d_{\bowtie}(f,g) = \infty$. Note however $d_1(f,g) = 2$. This can not happen for non-increasing and non-negative pcfs:

60. **Proposition.** *Assume f and g are non-increasing and non-negative pcfs. Then:*

(1) *$d_1(f,g) = \infty$ if and only if $d_{\bowtie}(f,g) = \infty$ if and only if $\lim(f) \neq \lim(g)$.*
(2) *$d_1(f,g) \leq \max\{f(0), g(0)\} d_{\bowtie}(f,g)$.*

Proof. (1): This is clear from the definitions and the fact that for a non-negative pcf h (such as $|f-g|$) $\lim(h) > 0$ if and only if $\int_{t=0}^{\infty} h = \infty$.
(2): If $d_{\bowtie}(f,g) = \infty$, the statement is clear. Assume f and g are ϵ interleaved. Consider $h = \max\{f,g\}$. The function $v \mapsto h(v+\epsilon)$, denoted by h_ϵ, is also a non-increasing and non-negative pcf. The Inequalities $f(v) \geq g(v+\epsilon)$ and $g(v) \geq f(v+\epsilon)$ together with the fact that f and g are non-increasing imply $h \geq f \geq h_\epsilon$ and $h \geq g \geq h_\epsilon$, and consequently $d_1(f,g) \leq d_1(h,h_\epsilon)$. Since:

$$d_1(h,h_\epsilon) = \int_0^\infty |h - h_\epsilon| = \int_0^\infty h - \int_0^\infty h_\epsilon = \int_0^\infty h - \int_\epsilon^\infty h = \int_0^\epsilon h \leq h(0)\epsilon$$

we can conclude $d_1(f,g) \leq \max\{f(0), g(0)\}\epsilon$. As this happens for any ϵ for which f and g are ϵ interleaved, we get the desired inequality. □

61. **Exercise.** Choose $a < b$ in $[0, \infty]$. Let f and g be pcfs. Define:

$$d(f,g) := \int_a^b |f - g|$$

Prove that d is a pseudo-metric on the set of pcfs. Show also that if $a \neq 0$ or $b \neq \infty$, then d is not a metric.

Extending distances to finite subsets

62. **Extensions.** Let X be a set. The symbol $\Delta[X]$ denotes the set of all non-empty finite subsets of X. We identify an element $x \in X$ with the corresponding singleton $\{x\}$ in $\Delta[X]$. Using this identification, we consider X as a subset of $\Delta[X]$. In this section we are going to discuss various ways of extending arbitrary distances on X to distances on $\Delta[X]$.

By restricting a distance on $\Delta[X]$ along the inclusion $X \subset \Delta[X]$, we obtain a distance on X (see Point 15). We think about this restriction procedure as a function $\mathrm{rest}\colon \mathcal{D}(\Delta[X]) \to \mathcal{D}(X)$. Any section to this restriction is called an **extension** along $X \subset \Delta[X]$. Thus an extension along $X \subset \Delta[X]$ is a function $\psi\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$, extending a distance d on X to a distance ψd on $\Delta[X]$, which makes the following diagram commutative:

$$\begin{array}{ccccc} \mathcal{D}(X) & \xrightarrow{\psi} & \mathcal{D}(\Delta[X]) & \xrightarrow{\mathrm{rest}} & \mathcal{D}(X) \\ & \searrow & & \nearrow & \\ & & \mathrm{id} & & \end{array}$$

Explicitly, $\psi\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$ is an extension if, for every distance d on X and all x and y in X:

$$d(x,y) = \psi d(\{x\},\{y\})$$

63. **Examples of extensions.** Let d be a distance on X. For non-empty finite subsets A and B of X define:

$$sd(A,B) := \min\{d(a,b) \mid a \in A \text{ and } b \in B\}$$

$$cd(A,B) := \begin{cases} \max\{d(a,b) \mid a \in A \text{ and } b \in B\} & \text{if } A \neq B \\ 0 & \text{if } A = B \end{cases}$$

$$ad(A,B) := \begin{cases} \frac{\sum_{a\in A, b\in B} d(a,b)}{|A|\ |B|} & \text{if } A \neq B \\ 0 & \text{if } A = B \end{cases}$$

Then sd, cd and ad are distances on $\Delta[X]$. In this way we defined three functions:

$$\mathcal{D}(X) \overset{\xrightarrow{s}}{\underset{\xrightarrow{a}}{\xrightarrow{c}}} \mathcal{D}(\Delta[X])$$

64. **Exercise.**

(1) Show that s, c and a are extensions along $X \subset \Delta[X]$. Explicitly, for every distance d on X and all x and y in X:

$$d(x,y) = sd(\{x\},\{y\}) = cd(\{x\},\{y\}) = ad(\{x\},\{y\})$$

(2) Show that if d is a pseudometric, then cd and ad are pseudometrics on $\Delta[X]$.
(3) Give an example of a pseudometric d for which sd is not a pseudometric.
(4) If d is a metric on X, are cd and ad also metrics on $\Delta[X]$?
(5) Give an example of a metric d on X for which sd is not a metric on $\Delta[X]$.

Comparing pseudometrics: fixed enumeration

It is important to be able to compare distance spaces to each other and evaluate differences between them. This is done by considering distances and (pseudo)metrics between distance spaces. One of the key aims of this course is to give various ways of producing such distances and (pseudo)metrics. To choose an appropriate such distance it is crucial to understand what constrains we need to respect. For example sometimes the underling set X should be fixed and sometimes we should allow it to change. Sometimes it is important to keep the order of the elements of X fixed and sometimes a weaker relation is desired when certain permutations of X should not change the distance to other distance spaces.

In this section we discuss distances in the case where we do not allow to change a fixed enumeration of X.

65. **Distances.** Let X be a set. Choose and fix an enumeration of its elements $X = \{x_1, \dots, x_k\}$. The space of distances $\mathcal{D}(X)$ on X can then be identified with symmetric $k \times k$ matrices with 0 diagonal (see Point 19). We can thus think about $\mathcal{D}(X)$ as a subset of $[0,\infty]^{k^2}$ and use all the (pseudo)metrics defined on $[0,\infty]^{k^2}$ in Points 45 and 49 as distances on $\mathcal{D}(X)$.

Since the distance matrices are symmetric with 0 diagonal, by disregarding elements on and above the diagonal, we could also think about $\mathcal{D}(X)$ as a subset $[0,\infty]^{\binom{k}{2}}$ (see Point 19). We can then use all the (pseudo)metrics defined on $[0,\infty]^{\binom{k}{2}}$ in Points 45 and 49 as distances on $\mathcal{D}(X)$.

66. **Exercise.** Choose a distance D on $\mathcal{D}(X)$ as described in Point 65. Let d and d' be discrete pseudometrics on X. Show that either $D(d,d') = 0$ or $D(d,d') = \infty$.

67. **Exercise.** Explain the relation between the corresponding distances on $\mathcal{D}(X)$ obtained by regarding $\mathcal{D}(X)$ as a subset of $[0,\infty]^{k^2}$ respectively as a subset of $[0,\infty]^{\binom{k}{2}}$.

68. **Invariance under permutations.** For a distance d on X and a permutation σ of X, define $d\sigma(x,y) := d(\sigma(x),\sigma(y))$ for x and y in X. We say that a distance D on $\mathcal{D}(X)$ is **invariant under permutations** if $D(d, d\sigma) = 0$ for all d in $\mathcal{D}(X)$ and all permutations σ of X.

Computational cost of calculating the distances on $\mathcal{D}(X)$ defined in Point 65 is low. That is an advantage. These distances however are not invariant under permutations of X and there are important situations in which this is a disadvantage.

69. **Exercise.** Give examples illustrating that all of the distances on $\mathcal{D}(X)$ defined in Point 45 are not invariant under permutations of X.

70. One way of defining distances on $\mathcal{D}(X)$ which are invariant under permutations is to first construct functions $f\colon \mathcal{D}(X) \to [0,\infty]$ which are invariant under permutations:

$f(d) = f(\sigma d)$ for any permutation σ of X. These function can be then used to form distances of the following form:

$$\mathcal{D}(X) \times \mathcal{D}(X) \xrightarrow{\begin{bmatrix} f \\ -f \end{bmatrix}} [0, \infty]$$

Comparing pseudometrics: Gromov-Hausdorff distance

71. Let (X, d_X) and (Y, d_Y) be distance spaces, $f\colon X \to Y$ a function, and ϵ be in $[0, \infty]$.

(1) f **preserves distances** if $d_X(x, y) = d_Y(f(x), f(y)$ for all x and y in X.
(2) f is **Lipschitz** if $d_X(x, y) \geq d_Y(f(x), f(y))$ for all x and y in X.
(3) f is an **embedding** if it is an inclusion and preserves distances.
(4) f is an **isometry** if it is a bijection and preserves distances.
(5) (X, d_X) and (Y, d_Y) are **isometric** if there exists an isometry between them.
(6) f is an ϵ **equivalence** if it is an embedding and, for every y in Y, there is x in X such that $d_Y(y, f(x)) \leq \epsilon$.

According to the definition, to verify if two pseudometric spaces are isometric we need to find a bijection preserving the pseudometrics. In general to find such a bijection we need in principle to investigate all the bijections. This is a very expensive operation.

72. **Exercise.** Show that the composition of functions preserves properties of being Lipschitz, being an isometry, being an embedding, and preserving distances. Give an example illustrating that being ϵ equivalence is not preserved by the composition.

73. **Exercise.** Show that (X, d) and (X, d') are isometric if and only if there is a permutation σ of X such that $d' = d\sigma$ (see Point 68).

74. **Exercise.** Show that if (X, d_X) and (Y, d_Y) are isometric pseudometric spaces, then both the partitions $(X, X/d_X)$ and $(Y, Y/d_Y)$ (see Exercise 11) and the metric spaces $(X/d_X, d_X)$ and $(Y/d_Y, d_Y)$ (see Point 40) are also isometric. Is the reverse implication also true? Construct an example.

75. **Distance glueing.** Consider embeddings $f\colon (A, d_A) \to (X, d_X)$ and $g\colon (A, d_A) \to (Y, d_Y)$. By identifying $f(a)$ with $g(a)$ in the disjoint union $X \coprod Y$, for every a in A, we obtain a set $X \cup_A Y$ called the **union** of X and Y along $f\colon A \to X$ and $g\colon A \to Y$. This union $X \cup_A Y$ fits into the following commutative square of functions:

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \downarrow \\ Y & \hookrightarrow & X \cup_A Y \end{array}$$

For pairs of elements x and y in the union $X \cup_A Y$ define:

$$d_{X \cup_A Y}(x, y) = \begin{cases} d_X(x, y) & \text{if } x, y \text{ are in } X \\ d_Y(x, y) & \text{if } x, y \text{ are in } Y \\ \inf\{d_X(x, f(a)) + d_Y(g(a), y) \mid a \in A\} & \text{if } \begin{array}{c} x \text{ is in } X \setminus f(A) \\ y \text{ is in } Y \setminus g(A) \end{array} \end{cases}$$

Let x and y be elements in $X \cup_A Y$ that belong to both $X \subset X \cup_A Y$ and $Y \subset X \cup_A Y$. This means that there are a and b in A such that $f(a) = x = g(a)$ and $f(b) = y = g(b)$. Since f and g are embeddings, $d_X(f(a), f(b)) = d_A(a, b) = d_Y(g(a), g(b))$, and hence $d_{X \cup_A Y}(x, y)$ is well defined.

The function $d_{X \cup_A Y}$ is a distance on $X \cup_A Y$ called **glueing** of (X, d_X) and (Y, d_Y) along $f\colon (A, d_A) \to (X, d_X)$ and $g\colon (A, d_A) \to (Y, d_Y)$.

76. **Exercise.** Show that the inclusions $X \subset X \cup_A Y$ and $Y \subset X \cup_A Y$ are embeddings.

77. **Exercise.** Show that if $f\colon (A, d_A) \to (X, d_X)$ and $g\colon (A, d_A) \to (Y, d_Y)$ are embeddings between pseudometric spaces, then the glueing $(X \cup_A Y, d_{X \cup_A Y})$ is also a pseudometric space. Is it true also for metric spaces?

78. **Proposition.**

(1) *Let $f\colon (A, d_A) \to (X, d_X)$ and $g\colon (A, d_A) \to (Y, d_Y)$ be embeddings. If f is an ϵ equivalence, then so is $Y \subset X \cup_A Y$.*
(2) *Assume Y is a pseudometric. If $f\colon A \to X$ is an ϵ equivalence and $h\colon X \to Y$ is a τ equivalence, then $hf\colon A \to Y$ is $\epsilon + \tau$ equivalence.*

Proof. (1): Let x be in $X \subset X \cup_A Y$. By the assumption, there is a in A such that $d_{X \cup_A Y}(x, f(a)) = d_X(x, f(a)) \leq \epsilon$. Since $f(a)$ is identified with $g(a)$, we get:

$$d_{X \cup_A Y}(x, g(a)) \leq d_X(x, f(a)) + d_Y(g(a), g(a)) = d_X(x, f(a)) \leq \epsilon$$

(2): Note that the assumption Y is a pseudometric implies that both A and X are also pseudometrics since f and h are embeddings. Clearly the composition hf is an embedding. Let y be in Y. Then there is x in X, such that $d_Y(h(x), y) \leq \tau$. We can then choose a in A so that $d_X(f(a), x) \leq \epsilon$. By the triangular inequality, we then obtain:

$$d_Y(hf(a), y) \leq d_Y(hf(a), h(x)) + d_Y(h(x), y) = d_X(f(a), x) + d_Y(h(x), y) \leq \epsilon + \tau$$

As this happens for any y in Y, we can conclude hf is an $\epsilon + \tau$ equivalence. □

79. **Exercise.** The pseudometric assumption in Proposition 78.(2) is crucial. Give an example of an ϵ equivalence $f\colon (A, d_A) \to (X, d_X)$ and a τ equivalence $g\colon (X, d_X) \to (Y, d_Y)$ of distance spaces for which the composition $gf\colon A \to Y$ is not an $\epsilon + \tau$ equivalence.

80. **Closeness and Gromov-Hausdorff distance.** Let ϵ be in $[0, \infty]$ and (X, d_X) and (Y, d_Y) be pseudometric spaces.

(1) (X, d_X) and (Y, d_Y) are called ϵ **close** if there is a pseudometric space (Z, d_Z), a τ_1 equivalence $f\colon X \to Z$, and a τ_2 equivalence $g\colon Y \to Z$ such that $\tau_1 + \tau_2 \leq \epsilon$.
(2) **Gromov-Hausdorff** distance between (X, d_X) and (Y, d_Y) is defined as:

$$\mathrm{GH}(X, Y) := \inf\{\epsilon \in [0, \infty] \mid (X, d_X) \text{ and } (Y, d_Y) \text{ are } \epsilon\text{-close}\}$$

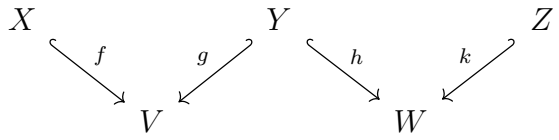
In the definition of ϵ closeness it is very important that we consider pseudometric spaces and not arbitrary distances. If we would allow embeddings into distance spaces, then according to the following Exercise the obtained distance would be always 0.

81**. Exercise.** Show that for all distance spaces (X, d_X) and (Y, d_Y), there is a distance space (Z, d_Z) and 0 equivalences $X \to Z$ and $Y \to Z$.

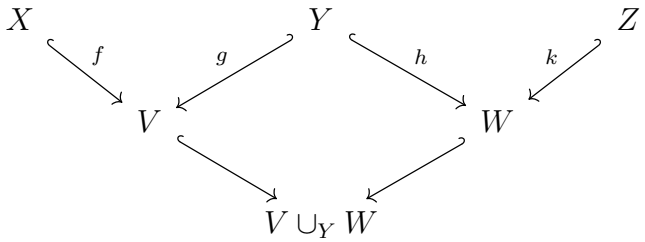
82**. Theorem.** *Gromov-Hausdorff distance satisfies the following properties:*

(1) *The Gromov-Hausdorff distance is a pseudometric on the collection of all pseudometric spaces.*
(2) *If X and Y are isometric, then* $\mathrm{GH}(X, Y) = 0$.
(3) *Let (X, d_X) and (Y, d_Y) be finite pseudometric spaces. Then* $\mathrm{GH}(X, Y) = 0$ *if and only if $(X/d_X, d_X)$ and $(Y/d_Y, d_Y)$ (see Point 40) are isometric.*

Proof. (1): The non-trivial part is to show the triangular inequality. Consider three pseudometric spaces (X, d_x), (Y, d_Y) and (Z, d_Z). By definition, for any real number $\epsilon > 0$, there exist the following isometric embeddings into some pseudometric spaces (V, d_V) and (W, d_W):



where f is τ_1 equivalence, g is τ_1 equivalence, h is τ_3-equivalence, k is τ_4-equivalence, $\tau_1 + \tau_2 < \mathrm{GH}(X, Y) + \epsilon$, and $\tau_3 + \tau_4 < \mathrm{GH}(Y, Z) + \epsilon$. Take the glueing of g and h and extend the above diagram to:



According to Proposition78.(1), $V \subset V \cup_Y W$ is τ_3 equivalence and $W \subset V \cup_Y W$ is τ_2 equivalence. Since $V \cup_Y W$ is a pseudometric (see Exercise 77), we can use Proposition 78.(2) to conclude that the composition $X \to V \cup_Y W$ is $\tau_1 + \tau_3$ equivalence and the composition $Z \to V \cup_Y W$ is $\tau_4 + \tau_2$ equivalence. This implies:

$$\mathrm{GH}(X, Z) \leq \tau_1 + \tau_3 + \tau_4 + \tau_2 \leq \mathrm{GH}(X, Y) + \epsilon + \mathrm{GH}(Y, Z) + \epsilon$$

As this happens for all real numbers $\epsilon > 0$, we get the triangular innequality:

$$\mathrm{GH}(X, Z) \leq \mathrm{GH}(X, Y) + \mathrm{GH}(Y, Z)$$

(2): This is a consequence of the fact that an isometry is a 0-equivalence.

(3): Let $f\colon X/d_X \to X$ and $g\colon Y/d_Y \to Y$ be any functions whose compositions with the quotient maps $X \to X/d_X$ and $Y \to Y/d_Y$ are the identity functions. Then f and g are 0 equivalences. If follows that $\mathrm{GH}(X, X/d_X) = \mathrm{GH}(Y, Y/d_Y) = 0$ and hence, since the Gromov-Hausdrof distance is a pseudometric $\mathrm{GH}(X, Y) = \mathrm{GH}(X/d_X, Y/d_Y)$. Thus, according to statement (2), if X/d_X and Y/d_Y are isometric, then $\mathrm{GH}(X, Y) = 0$.

The quotients X/d_X and Y/d_Y are metrics (see Proposition 41). Considering that $GH(X, Y) = GH(X/d_X, Y, d_Y)$, it thus remain to be show that under the assumption (X, d_X) and (Y, d_Y) are metrics, vanishing of the Gromov-Hausdorff distance $\mathrm{GH}(X, Y) = 0$ implies X and Y are isometric.

Assume (X, d_X) and (Y, d_Y) are metrics and $\mathrm{GH}(X, Y) = 0$. Since X and Y are finite, there is a real number $\epsilon > 0$ which is smaller than any positive value of the distance functions d_X and d_Y. Let $f\colon X \to Z$ and $g\colon Y \to Z$ be τ_1 respectively τ_2 equivalences such that $\tau_1 + \tau_2 < \epsilon/2$. Choose x in X. Since g is $\tau_2 < \epsilon/2$ equivalence, there is y in Y such that $d_Z(f(x), g(y)) \leq \tau_2$. We claim that such a y is unique. If there were two such elements y_1 and y_2, then:

$$d_Y(y_1, y_2) = d_Z(g(y_1), g(y_2)) \leq d_Z(g(y_1), f(x)) + d_Z(f(x), g(y_2)) \leq \tau_2 + \tau_2 < \epsilon$$

and consequently $d_Y(y_1, y_2) = 0$ as d_Y has no positive values smaller than ϵ. Since Y is a metric, $y_1 = y_2$.

By mapping x to y we obtain an embedding $X \to Y$. By an analogous symmetric argument we also obtain an embedding $Y \to X$. These embeddings are the desired isometries. □

83**. Exercise.** Let (X, d_X) be a discrete pseudometric. Show that if (Y, d_Y) is a pseudometric such that $\mathrm{GH}(X, Y) = 0$, then d_Y is discrete.

Kleinberg theorem

84. Let X be a set. If d is a pseudometric on X, then the function $\alpha d\colon X \times X \to [0, \infty]$ is also a pseudometric for any α in $[0, \infty)$. Furthermore, for any α in $(0, \infty)$, d is a metric if and only if αd is a metric.

85. **Exercise.** Prove the above statement.

86. Let X be a set and d_0 a discrete pseudometric on X.

(1) We say that a pseudometric d' on X is d_0-**transform** of a pseudometric d if:
- $d'(x, y) \leq d(x, y)$ when $d_0(x, y) = 0$;
- $d'(x, y) \geq d(x, y)$ when $d_0(x, y) = \infty$.

(2) Let $0 < a < b$ be positive real numbers. We say that a pseudometric d on X is d_0-**transform** of the pair (a, b) if:
- $d(x, y) \leq a$ when $d_0(x, y) = 0$;
- $d(x, y) \geq b$ when $d_0(x, y) = \infty$.

Note that d_0 is always the d_0-transform of any metric.

87. Recall that discrete pseudometrics on X are identified with partitions. Let us reformulate the definitions presented in 86 in terms of a partition P.

(1) a pseudometric d' on X is P-**transform** of a pseudometric d if:
- $d'(x, y) \leq d(x, y)$ when x and y are in the same block of P;
- $d'(x, y) \geq d(x, y)$ when x and y are not in the same block of P.

(2) Let $0 < a < b$ be positive real numbers. We say that a pseudometric d on X is P-**transform** of the pair (a, b) if:
- $d(x, y) \leq a$ when x and y in the same block of P;
- $d(x, y) \geq b$ when x and y are not in the same block of P.

We think of a P-transform of a pseudometric d as a pseudometric d' whose distances on blocks of P can not be bigger than distances given by d, however distances of d' between points from different blocks should be at least as big as the distances given by d. Thus a P-transform of d is its modification that preserves or decreases distances between points in the same blocks of P and preserves or increases distances between points in different blocks of the partition.

Similarly for a P-transform of a pair of numbers (a, b). For points in the same block of P the distances of such a transform should not be bigger than a, but for points in different blocks, the distances should be at least as big as b.

88. Since general pseudometrics are difficult to visualise and comprehend, one strategy is to simplify them. As discrete pseudometrics are one of the easiest to understand, we

might look for simplifying functions of the form:

$$\mathcal{P}M(X) \xrightarrow{\quad C \quad} \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$$

Arbitrary functions are not very useful. Useful simplifying functions should preserve certain properties. Here are some desirable features that we would like such a simplifying function to satisfy:

- **Scale Invariance.** For any pseudometric d and any positive real number α, $C(d) = C(\alpha d)$ (note that if d is a pseudometric on X, then so is αd).
- **Richness.** If d is discrete, then $C(d) = d$.
- **Consistency.** If d' is a $C(d)$-transform of d, then $C(d') = C(d)$.

Why do we care about such functions? For example let X be a set of patients and let a pseudometric on X be given by differences between some biological measurements taken from subjects in X. Such measurements would form finite subsets of euclidean spaces and to calculate differences between measurements taken from different subjects we might use some of the metrics given in Point 45. In such a scenario often the aim is to use these biological measurements and the differences between them to retrieve clinical information about the subjects for example who is a case and who is a control. This would indicate a relation between biological information and the fact that a subject is a case. Such an analysis amounts to translating information encoded by the measurements on X (the pseudometric) into a partition of X. Avoiding overfitting issues it is more proper to use generic translations and not translations catered specifically to this one pseudometric. Such a generic translation can be regarded as a function $C\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ describing how measurements lead to partitions (recall that discrete pseudometrics and partitions on X are identified). Of course changing the units of the measurements, should not affect if a given subject is a case or a control. Thus the **scale invariance** is a desirable property for such a translation.

If the pseudometric is already discrete, then it is not an unreasonable requirement that such a pseudometric is translated into itself, as there is no other information a discrete pseudometric contains. **Richness** is therefore also a desirable property.

Consistency is the most difficult to justify. It states that if a given pseudometric leads to a particular partition, then if we transform this metric by possibly decreasing distances between points in the same block of this partition and possibly increasing distances between points in different blocks, then such a transformed pseudometric should lead to the same partition.

89**.** Recall that we have already defined a function $L{-}_t\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ from the set of pseudometrics on X to the set of discrete pseudometrics on X (see Points 21 and 29). This function however is not scale invariant:

Proposition. *For any t in $\mathbf{R}_{\geq 0}$, the function $L{-}_t\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ is rich and consistent but not scale invariant.*

Proof. If d is discrete, then $Ld_t = d$ which is richness. Assume that d' is Ld_t transform of d. We need to show that $Ld'_t = Ld_t$. If $Ld_t(x,z) = 0$, then there is a sequence $x = z_0, z_1, \ldots, z_n = y$ such that $d_t(z_i, z_{i+1}) = 0$ and hence $d(z_i, z_{i+1}) \leq t$. It follows that $d'(z_i, z_{i+1}) \leq d(z_i, z_{i+1}) \leq t$ and therefore $Ld'_t(x,z) = 0$. A similar argument shows that if $Ld_t(x,z) = \infty$, then so $Ld'_t(x,z) = \infty$. □

90. Choose a natural number k such that $k \leq |X|$. This assumption implies that the following set is non-empty:

$$\{t \in [0,\infty) | \text{ the size of the partition } X/d_t \text{ (see Point 25) is at least } k\}$$

For $k > 1$, this set is also bounded since X is finite. For a pseudometric d on X, define:

$$t_k := \sup\{t \mid |X/d_t| \geq k\}$$

Proposition. *For a fixed k, the function $L-_{t_k}\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ is scale invariant and consistent but not rich.*

91. **Exercise.** Prove the above proposition.

92. **Theorem.** (Kleinberg). *If $|X| \geq 2$, then there is no function $C\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ that is scale invariant, rich, and consistent.*

Proof. The theorem follows from the following two lemmas and the fact that there are more than 1 discrete pseudometrics on X if $|X| \geq 2$. □

93. **Lemma.** *Assume $C\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ is consistent. Then there are positive real numbers $a < b$ such that any $C(d)$-transform of (a,b) is mapped via C to $C(d)$.*

Proof. Consider the following sets of real numbers:

$$M_1 := \{d(x,y) \mid \text{ for } x \text{ and } y \text{ such that } d(x,y) > 0 \text{ and } C(d)(x,y) = 0\}$$

$$M_2 := \{d(x,y) \mid \text{ for } x \text{ and } y \text{ such that } d(x,y) > 0 \text{ and } C(d)(x,y) = \infty\}$$

Define $a' := \min M_1$ and $b' := \max M_2$ and choose $0 < a < b$ so that $a < a'$ and $b' < b$. Note that any $C(d)$-transform of (a,b) is a $C(d)$-transform of d. Hence by consistency, C maps any such pseudometric to $C(d)$. □

94. **Lemma.** *Assume $C\colon \mathcal{P}M(X) \to \mathrm{Dis}\mathcal{P}\mathcal{M}(X)$ is consistent and scale invariant. Let d_0 and d_1 be pseudometrics on X for which the discrete pseudometric $C(d_0)$ is a submetric of the discrete pseudometric $C(d_1)$ (this means that if $C(d_0)(x,y) = 0$, then $C(d_1)(x,y) = 0$). Then $C(d_0) = C(d_1)$.*

Proof. Use Lemma 93 to obtain positive real numbers $0 < a_0 < b_0$ and $0 < a_1 < b_1$ such that, any $C(d_i)$-transform of (a_i, b_i) is mapped via C to $C(d_i)$ for $i = 0, 1$. Choose a_2 so that $0 < a_2 < a_1$ and ϵ so that $0 < \epsilon < a_0 a_2 / b_0$. Choose a pseudometric $b\colon X \times X \to [0,\infty]$ on X that satisfies the following properties:

- $b(x,y) < \epsilon$ if $C(d_0)(x,y) = 0$

- $a_2 \leq b(x,y) \leq a_1$ if $C(d_0)(x,y) = \infty$ and $C(d_1)(x,y) = 0$
- $b(x,y) \geq b_1$ if $C(d_0)(x,y) = \infty$ and $C(d_1)(x,y) = \infty$.

Note that b is a pseudo-metric which is a $C(d_1)$- transform of (a_1, b_1) and hence $C(b) = C(d_1)$.

Let $\alpha = b_0/a_2$ and consider the pseudometric αb. Then:

- $\alpha b(x,y) \leq a_0$ if $C(d_0)(x,y) = 0$
- $\alpha b(x,y) \geq b_0$ if $C(d_0)(x,y) = \infty$.

This means that $\alpha b(x,y)$ is $C(d_0)$-transform of (a_0, b_0). We can then conclude that

$$C(d_0) = C(\alpha b(x,y)) = C(b) = C(d_1)$$

□

Non Archimedean distances

One aspect of this course is to present tools to visualise and describe pseudometrics in ways that can be helpful to comprehend and understand them. For example in Proposition 35 it was explained how discrete pseudometrics can be identified with partitions. These are the easiest pseudometrics to visualise and understand. In general we should not expect to be able to identify in a similar way arbitrary pseudometrics with basic combinatorial structures. Later in the course we explain how arbitrary pseudometrics can be identified with objects constructed out of simplicial complexes illustrating the fact that in general it is necessary to use complicated topological constructions. One might wonder however if there is a class of pseudometrics which are more general than discrete ones and that still can be identified with elementary combinatorial structures. The aim of this section is to introduce such pseudometrics which are called non Archimedean distances or ultra pseudometrics. We are going to show that such pseudometrics can be identified with so-called dendrograms in an analogous way to how discrete pseudometrics have been identified with partitions.

95**.** A distance d on a set X is called an **non Archimedean** if it satisfies the following condition for all x, y and z in X:

(3)' $\max\{d(x,y),d(y,z)\} \geq d(x,z)$.

The symbol $\mathrm{Ultra}(X)$ denotes the set of all non Archimedean distances on X.

Since $d(x,y)+d(y,z) \geq \max\{d(x,y),d(y,z)\}$, a non Archimedean distance satisfies the triangular inequality and hence it is a pseudometric. We thus have an inclusion $\mathrm{Ultra}(X) \subset \mathcal{PM}(X)$. A non Archimedean distance is also called an **ultra** pseudometric.

A discrete pseudometric is also an ultra pseudometric and hence we have an inclusion $\mathrm{Dis}\mathcal{PM}(X) \subset \mathrm{Ultra}(X)$.

96**. Proposition.** *A distance d on X is an ultra pseudometric if and only if, for any 3 points x,y, and z in X, one of the following conditions is satisfied:*

- $d(x,y) = d(x,z) \geq d(y,z)$ *or*
- $d(y,x) = d(y,z) \geq d(x,z)$ *or*
- $d(z,x) = d(z,y) \geq d(x,y)$

According to this proposition an ultra pseudometric is a distance for which all the triangles xyz are acute and isosceles (two of its sides have the same length).

Proof. Let d be an ultra pseudometric on X. Choose 3 points x, y and z in X and order them such that $d(x,z) \geq d(x,y) \geq d(y,z)$. By definition $\max\{d(x,y),d(y,z)\} \geq d(x,z)$, which gives the equality $d(x,z) = d(x,y)$. Consequently the triangle is acute and isosceles. The reverse implication is clear. □

The fact that all triangles are acute and isosceles in an ultra pseudometric space has unusual consequences challenging our intuition coming from triangles, balls and disks (see Point 18) in the euclidean spaces. For example:

97. **Proposition.** *Let d be an ultra pseudometric on X. Then:*

(1) *If $d(x,y) < r$, then $B(x,r) = B(y,r)$.*
(2) *If $d(x,y) \leq r$, then $D(x,r) = D(y,r)$.*
(3) *If $B(x,r) \cap B(y,s) \neq \emptyset$, then either $B(x,r) \subset B(y,s)$ or $B(y,s) \subset B(x,r)$.*
(4) *If $d(x,y) \leq t$ and $d(y,z) \leq t$, then $d(x,z) \leq t$.*
(5) *The following are equivalent:*
   - *$d(x,y) \leq t$*
   - *there is a sequence $x = z_0, z_1, \ldots, z_n = y$ such that $d(z_i, z_{i+1}) \leq t$.*

Proof. (1) and (2): Since the arguments for (1) and (2) are identical, we present only how to show (1).

If $d(x,y) < r$, then for any z with $d(x,z) < r$, since $\max\{d(x,y), d(x,z)\} \geq d(y,z)$ we have $d(y,z) < r$. Consequently $B(x,r) \subset B(y,r)$. Symmetric argument gives $B(y,r) \subset B(x,r)$, and hence we get an equality $B(x,r) = B(y,r)$.

(3): Assume $r \leq s$. Choose z in $B(x,r) \cap B(y,s)$. This means $d(x,z) < r$ and $d(y,z) < s$. We can then use (1) to conclude $B(x,r) = B(z,r) \subset B(z,s) = B(y,s)$

(4): This is a direct consequence of (1).

(5): This follows from (4). □

98. **From ultra pseudometrics to discrete pseduometrics.** Let t be in $[0,\infty)$. In Point 29, we described how an arbitrary distance space (X,d) can be converted into a discrete distance space (X,d_t):

$$d_t(x,y) := \begin{cases} 0 & \text{if } d(x,y) \leq t \\ \infty & \text{otherwise} \end{cases}$$

One might ask under what circumstances this distance is a pseudometric (necessary discrete)? The answer is provided by:

99. **Proposition.** *Let d be a distance on X. The function d_t is a pseudometric for all t in $[0,\infty)$ if and only if d is an ultra pseudometric.*

Proof. d_t is a pseudometric if and only if it satisfies the triangular inequality $d_t(x,y) + d_t(y,z) \geq d_t(x,z)$ for any x, y, z in X. We need to show that this happens for all t if and only if d is an ultra pseudometric.

Assume d is an ultra pseudometric. If $d(x,y) > t$ or $d(y,z) > t$, then since $\infty \geq r$ for any r in $[0,\infty]$, the required triangular inequality holds. Let $d(x,y) \leq t$ and $d(y,z) \leq t$. Then since d is an ultra pseudometric, by Proposition 97.(4), also $d(x,z) \leq t$. The required triangular inequality for d_t holds therefore in this case as well.

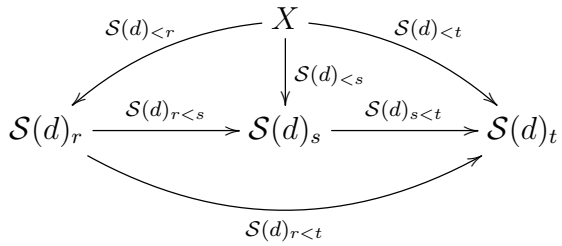
Assume that the required triangular inequalities hold. Choose x, y, z in X. Assume $d(x,y) \leq d(y,z)$. Then for $t = d(y,z)$, since $0 = d_t(x,y) + d_t(y,z) \geq d_t(x,z)$ we can

conclude that also $d(x,z) \leq t = d(y,z)$. Consequently $\max\{d(x,y), d(y,z)\} \geq d(x,z)$ and so d is an ultra pseudometric. $\square$

100**. Single linkage.** For any pseudometric d on X (ultra or not) the function d_0 is always a discrete pseudometric. It corresponds to the partition $\mathcal{S}(d)_0 := X/d_0$ of X that consists of subsets of X of the form $[x]_0 = \{y \mid d(x,y) = 0\}$ for various x in X.

If d is an ultra pseudometric on X, then, for any t in $[0,\infty)$, d_t is also a discrete pseudometric. It corresponds to the partition $\mathcal{S}(d)_t := X/d_t$ of X that consists of subsets of X of the form $[x]_t := \{y \mid d(x,y) \leq t\}$ for various x in X.

Note that if $s < t$, then the partition $\mathcal{S}(d)_s$ is a sub-partition of $\mathcal{S}(d)_t$ (see Point 9). Define $\mathcal{S}(d)_{s<t} \colon \mathcal{S}(d)_s \to \mathcal{S}(d)_t$ to be the function induced by $\mathrm{id} \colon X \to X$. It maps a block $[x]_s \subset X$ in $\mathcal{S}(d)_s$ to the block $[x]_t \subset X$ in $\mathcal{S}(d)_t$. Denote also by the symbol $\mathcal{S}(d)_{<t} \colon X \to \mathcal{S}(d)_t$ the quotient function mapping x to the block $[x]_t$ containing it. It is then clear that, for any $r < s < t$ in $[0,\infty)$, the following diagram of functions of sets commutes:



Note further that the functions $\mathcal{S}(d)_{<t} \colon X \to \mathcal{S}(d)_t$, and consequently $\mathcal{S}(d)_{s<t} \colon \mathcal{S}(d)_s \to \mathcal{S}(d)_t$, for all $s < t$, are surjective.

These functions have another property in case X is a finite set. Assume X is finite. Although we have defined $\mathcal{S}(d)_{s<t}$ for all pairs of non-negative real numbers $s < t$, most of these functions are in fact just the identities. To see this, since X finite, $d \colon X \times X \to [0,\infty]$ has only finitely many values. We can therefore list these values in an increasing order and augment, if necessary, by 0 and ∞ to get a sequence:

$$0 = a_0 < a_1 < \cdots < a_n < \infty$$

Note that if both $s < t$ belong to the same interval among $[a_0, a_1), \ldots, [a_{n-1}, a_n), [a_n, \infty)$, then $d_s = d_t$ and consequently the map $\mathcal{S}(d)_{s<t} \colon \mathcal{S}(d)_s \to \mathcal{S}(d)_t$ is the identity.

We are going to use the above observations to define a new structure on a finite set which is called a dendrogram.

Dendrograms

101. A **tame set parametrized by** $[0,\infty)$ is a sequence of sets D_t, indexed by t in $[0,\infty)$, together with a sequence of functions of sets $D_{s<t}\colon D_s \to D_t$, called **transitions**, indexed by pairs $s<t$ of elements in $[0,\infty)$. These functions are required to satisfy the following properties:

(1) for any $r<s<t$, the following diagram commutes:

$$\begin{array}{ccccc} D_r & \xrightarrow{D_{r<s}} & D_s & \xrightarrow{D_{s<t}} & D_t \\ & \searrow & & \nearrow & \\ & & D_{r<t} & & \end{array}$$

(2) there is a sequence of real numbers $0=a_0<a_1<\cdots<a_n$ such that $D_{s<t}\colon D_s \to D_t$ is an isomorphism for any pair $s<t$ for which both s and t belong to the same interval among $[a_0,a_1),\ldots,[a_{n-1},a_n),[a_n,\infty)$. We say that the sequence $0=a_0<a_1<\cdots<a_n$ **discretises** the tame parametrised set.

A tame set D parametrized by $[0,\infty)$ is called a **dendrogram** if in addition:

(3) the function $D_{s<t}\colon D_s \to D_t$ is surjective for all $s<t$ in $[0,\infty)$.

Let X be a set. Then the same symbol X denotes also the **constant dendrogram** which by definition consists of $X_t=X$ for all t in $[0,\infty)$, with the transition function $X_{s<t}\colon X\to X$ given by the identity $\mathrm{id}\colon X\to X$ for all $s<t$.

102. **Maps.** Let D and D' be tame sets parametrized by $[0,\infty)$. A **map** between D and D', denoted by $f\colon D\to D'$, is defined to be a sequence of functions $f_t\colon D_t\to D'_t$, for all t in $[0,\infty)$, for which the following diagram commutes for all $s<t$ in $[0,\infty)$:

$$\begin{array}{ccc} D_s & \xrightarrow{D_{s<t}} & D_t \\ {\scriptstyle f_s}\downarrow & & \downarrow{\scriptstyle f_d} \\ D'_s & \xrightarrow{D'_{s<t}} & D'_t \end{array}$$

Such a map is:

(1) an **isomorphism** if f_t is a bijection for all t;
(2) a **surjection** if f_t is a surjection for all t;
(3) an **inclusion** if f_t is an inclusion for all t.

103. **Exercise.** Let X and Y be sets. Show that the set of maps between the constant dendrograms X and Y is in bijection with the set of functions between the sets X and Y.

104**. Exercise.** Show that a tame set D parametrized by $[0,\infty)$ is isomorphic to a constant dendrogram if and only if the transition $D_{s<t}\colon D_s \to D_t$ is a bijection for all $s<t$ in $[0,\infty)$.

105**. Dendrograms on X.** Let X be a set. A map $X \to D$, called the **structure map**, between the constant dendrogram X and a tame set D parametrized by $[0,\infty)$, is also called a **tame set parametrized by** $[0,\infty)$ **on** X. Such a map $X \to D$ is called a **dendrogram on** X if it is a surjection, i.e., if the structure functions $X \to D_t$ are surjections for all t in $[0,\infty)$.

If D is a tame set parametrized by $[0,\infty)$ on X, then we use the same symbol D to denote the range of the structure map and the symbol $D_{<t}\colon X \to D_t$ to denote the structure map at t.

The symbol $\mathrm{Dend}(X)$ denotes the set of all dendrograms on X.

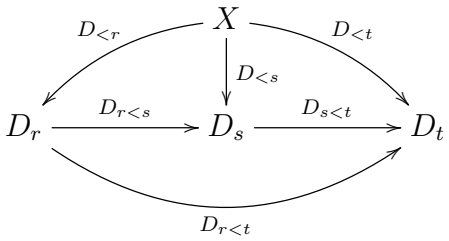
A tame set $X \to D$ parametrized by $[0,\infty)$ on X is called **discrete** if D is isomorphic to a constant dendrogram. According to Exercise 104, $X \to D$ is discrete if $D_{s<t}\colon D_s \to D_t$ is a bijection for all $s<t$.

Explicitly, a tame set D parametrized by $[0,\infty)$ on X is a sequence of sets D_t, indexed by t in $[0,\infty)$, together with two sequences of functions of sets:

- $D_{<t}\colon X \to D_t$, indexed by t in $[0,\infty)$
- $D_{s<t}\colon D_s \to D_t$, indexed by pairs $s<t$ of elements in $[0,\infty)$.

These functions are required to satisfy the following properties:

(1) for any $r<s<t$, the following diagram of functions of sets commutes:



(2) there is a sequence of real numbers $0 = a_0 < a_1 < \cdots < a_n$ such that $D_{s<t}\colon D_s \to D_t$ is the identity map for any pair $s<t$ for which both s and t belong to the same interval among $[a_0,a_1),\ldots,[a_{n-1},a_n),[a_n,\infty)$.

If in addition:

(3) the map $f_t\colon X \to D_t$ is surjective for any t in $[0,\infty)$,

then such a tame set parametrized by $[0,\infty)$ on X is a dendrogram on X.

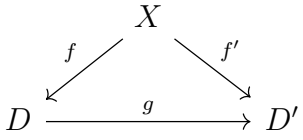
For example in Point 100 we descibed how an ultra pseudometric d on X leads to a dendrogram $\mathcal{S}(d)$ on X.

Note that to describe a dendrogram D on X, it is enough to specify the surjections $D_{<t}\colon X \to D_t$ for all t. Once these functions are given, since they are surjective, we have no choice for the functions $D_{s<t}\colon D_s \to D_t$ which also have to be surjective.

Keep in mind however that not all sequences of surjections $\{X \to D_t\}_{t\in[0,\infty)}$ lead to a dendrogram on X.

106**. Exercise.** Let $X \to D$ be a dendrogram on X. Show that D is a dendrogram.

107**.** A **map under** X between two tame sets $f\colon X \to D$ and $f'\colon X \to D'$ parametrized by $[0,\infty)$ on X is by definition a map $g\colon D \to D'$ making the following diagram commutative:



Such a map is an isomorphism if g is an isomorphism.

108**. Exercise.** Let D and D' be dendrograms on X. Show that there exists at most only one map $g\colon D \to D'$ under X.

109**. Constructing dendrograms on** X**.** Let D be a dendrogram on X. We can think about the function $D_{<t}\colon X \to D_t$ as a quotient (see Point 6). Recall that such quotient functions are identified with partitions (see Point 38). Thus a dendrogram on X can be also described as a collection of partitions P_t of X, indexed by t in $[0,\infty)$, such that:

- for $s < t$, the partition P_s is a sub partiton of P_t;
- there is a sequence $0 = a_0 < a_1 < \cdots < a_n < \infty$ of real numbers such that, if both $s < t$ belong to the same interval among $[a_0, a_1), \ldots, [a_{n-1}, a_n), [a_n, \infty)$, then $P_s = P_t$.

Partitions and quotients of X are also identified with discrete pseudometrics on X (see Point 38). If we use this identification, then a dendrogram on X can be described also as a collection of discrete pseudometrics d_t on X, indexed by t in $[0,\infty)$, such that:

- for $s < t$, if $d_s(x, y) = 0$, then $d_t(x, y) = 0$;
- there is a sequence $0 = a_0 < a_1 < \cdots < a_n < \infty$ of real numbers such that, if both $s < t$ belong to the same interval among $[a_0, a_1), \ldots, [a_{n-1}, a_n), [a_n, \infty)$, then $d_s = d_t$.

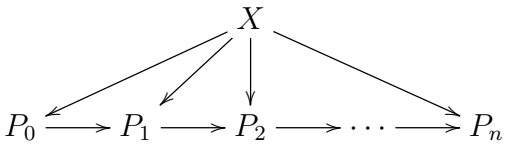
To define a dendrogram on a set X we can use a table of the form:

real numbers	$0 = a_0$	a_1	a_2	$\cdots$
partitions on X	P_0	P_1	P_2	$\cdots$
discrete pseudometrics on X	d_0	d_1	d_2	$\cdots$
quotients of X	$X \to P_0$	$X \to P_1$	$X \to P_2$	$\cdots$

where

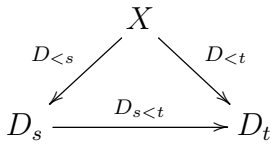
- the first row is a finite sequence of real numbers $0 = a_0 < a_1 < \cdots < a_n$;
- the second row is a finite sequence of structures on X equivalently described by either:

– partitions of X such that $P_0 \geq P_1 \geq \cdots \geq P_n$;
– discrete pseudometrics on X such that $d_0 \geq d_1 \geq \cdots \geq d_n$;
– quotients of X for which there are functions $P_i \to P_{i+1}$ (necessary unique and surjective) making the following diagram commutative:



Here is how such a table leads to a dendrogram D on X:

- For $t \in [a_i, a_i + 1)$ define $D_t := P_i$ or $D_t := X/d_i$, depending on the choice of the second row.
- For $t \in [a_i, a_i + 1)$ set $D_{<t} \colon X \to D_t$ to be the quotient function $X \to P_i$ or $X \to X/d_i$, again depending on the choice of the second row.
- for $0 \leq s < t$, set $D_{s<t} \colon D_s \to D_t$ to be the unique function that makes the following diagram commutative:

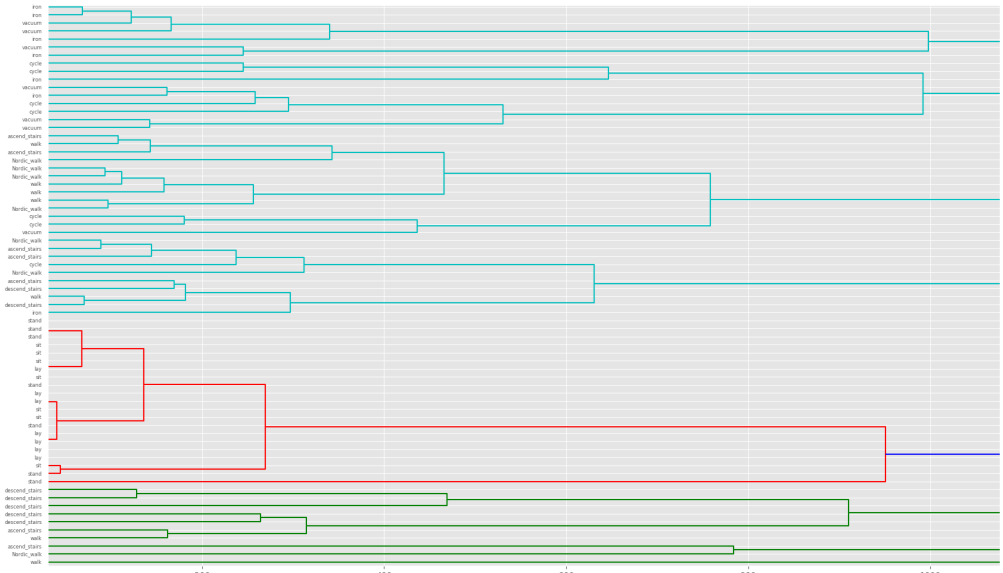


Existence of such functions follows from the assumptions on the sequence in the second row.

110**. Exercise.** Show that a dendrogram D on X is discrete if and only if it is isomorphic to a dendrogram given by a table of the form:

real numbers	$0 = a_0$
partitions on X	P_0
discrete pseudometrics on X	d_0
quotients of X	$X \to P_0$

111**. Visualisation.** A key advantage of dendrograms is that they can be graphically visualized in a way which can be helpful in extracting and comprehending their properties. Here is an example of such a visualization:


iron
iron
vacuum
vacuum
iron
vacuum
iron
cycle
cycle
iron
vacuum
iron
cycle
cycle
vacuum
vacuum
ascend_stairs
walk
ascend_stairs
Nordic_walk
Nordic_walk
Nordic_walk
walk
walk
walk
Nordic_walk
cycle
cycle
vacuum
Nordic_walk
ascend_stairs
ascend_stairs
cycle
Nordic_walk
ascend_stairs
descend_stairs
walk
descend_stairs
iron
stand
stand
stand
sit
sit
sit
lay
sit
stand
lay
lay
sit
sit
stand
lay
lay
lay
lay
sit
stand
stand
descend_stairs
descend_stairs
descend_stairs
descend_stairs
descend_stairs
ascend_stairs
walk
ascend_stairs
Nordic_walk
walk
200
400
600
800
1000

Dendrograms and ultra pseudometrics

The aim of this section is to explain how to identify ultra pseudometrics on X with dendrograms on X.

112. In Point 100 we described how an ultra pseudometric d on a finite set X leads to a dendrogram $\mathcal{S}(d)$ on X. Recall that $\mathcal{S}(d)_{<t}\colon X \to \mathcal{S}(d)_t$ maps an element x in X to its block $[x]_t$ in $\mathcal{S}(d)_t = X/d_t$ which consists of all elements y in X such that $d(x,y) \leq t$. By mapping d to $\mathcal{S}(d)$ we obtain a function:

$$\mathcal{S}\colon \mathrm{Ultra}(X) \to \mathrm{Dend}(X)$$

We can reverse this process. Let D be a dendrogram on a set X. Define $u(D)\colon X \times X \to [0,\infty]$ as follows:

$$u(D)(x,y) := \inf\{t \in [0,\infty] \mid D_{<t}(x) = D_{<t}(y)\}$$

113. **Proposition.** *$u(D)\colon X \times X \to [0,\infty]$ is an ultra pseudometric on X.*

Proof. We start with a lemma: *If $s := u(D)(x,y)$, then $D_{<s}(x) = D_{<s}(y)$.*

The lemma assures that the infimum $u(D)(x,y) = \inf\{t \in [0,\infty] \mid D_{<t}(x) = D_{<t}(y)\}$ belongs to the set $\{t \in [0,\infty] \mid D_{<t}(x) = D_{<t}(y)\}$.

To prove this lemma choose a sequence $0 = a_0 < a_1 < \cdots < a_n < \infty$ that discretises D. Then there is $\epsilon > 0$ in $[0,\infty)$ such that $[s, s+\epsilon)$ is included entirely in one of the intervals $[a_0,a_1), \ldots, [a_{n-1},a_n), [a_n,\infty)$. The function $D_{s<s+\epsilon}\colon D_s \to D_{s+\epsilon}$ is therefore the identity and since $D_{<s+\epsilon}(x) = D_{<s+\epsilon}(y)$, then also $D_{<s}(x) = D_{<s}(y)$.

Let x, y, z be elements in X. Set $s := \max\{u(D)(x,y), u(D)(y,z)\}$. By the Lemma $D_{<s}(x) = D_{<s}(y) = D_{<s}(z)$ and therefore:

$$s = \max\{u(D)(x,y), u(D)(y,z)\} \geq u(D)(x,z)$$

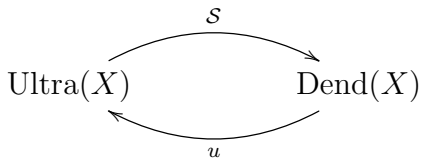
This proves $u(D)$ is an ultra pseudometric on X. □

In this way we obtained a function:

$$u\colon \mathrm{Dend}(X) \to \mathrm{Ultra}(X)$$

which maps a dendrogram D on X to an ultra pseudometric $u(D)$ on X.

114. **Proposition.** *The functions:*

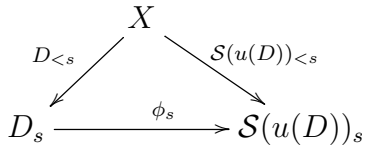


satisfy the following properties:

(1) *For any ultra pseudometric d on X, $u(\mathcal{S}(d)) = d$.*
(2) *For any dendrogram D on X, $\mathcal{S}(u(D))$ and D are isomorphic.*

Proof. (1): Assume $d(x, y) \leq t$. Consider the function $\mathcal{S}(d)_{<t} \colon X \to \mathcal{S}(d)_t$. Since $d(x, y) \leq t$, we have $\mathcal{S}(d)_{<t}(x) = \mathcal{S}(d)_{<t}(y)$. This implies $u(\mathcal{S}(d))(x, y) \leq t$. Consequently $d(x, y) \geq u(\mathcal{S}(d))(x, y)$. To show the reverse inequality, assume $u(\mathcal{S}(d))(x, y) \leq t$ which implies $\mathcal{S}(d)_{<t}(x) = \mathcal{S}(d)_{<t}(y)$. The blocks $[x]_t$ and $[y]_t$ of the the partition $\mathcal{S}(d)_t$ therefore coincide, and hence $d(x, y) \leq t$ which proves $d(x, y) \leq u(\mathcal{S}(d))(x, y)$.

(2): Consider the functions $D_{<s} \colon X \to D_s$ and $\mathcal{S}(u(D))_{<s} \colon X \to \mathcal{S}(u(D))_s$. By definition $\mathcal{S}(u(D))_{<s}(x) = \mathcal{S}(u(D))_{<s}(y)$ if and only if $d_D(x, y) \leq s$. This happens if and only if $D_{<s}(x) = D_{<s}(y)$. Thus two elements of X are mapped to the same element via $D_{<s}$ if and only if they are mapped to the same element via $\mathcal{S}(u(D))_{<s}$. This means that there is a bijection $\phi_s \colon D_s \to \mathcal{S}(u(D))_s$ for which the following diagram commutes, proving the statement:



□

115**.** From now on we identify ultra pseudometrics on X with isomorphism classes of dendrograms on X via the functions u and $\mathcal{S}$ described above. Dendrograms on X provide a convenient and useful way of visualising ultra pseudometrics on X.

116**. Exercise.** Show that via the identifications u and $\mathcal{S}$, discrete dendrograms are identified with discrete pseudometrics on X.

Hierarchical clustering

The aim of this section is to explain how to construct various extensions of the function $\mathcal{S}\colon \mathrm{Ultra}(X) \to \mathrm{Dend}(X)$ (see point 112) to the set $\mathcal{D}(X)$ of all distances on X.

117**. Constructing dendrograms.** Let X be a set. Recall that $\Delta[X]$ denotes the set of all finite non-empty subsbets of X. Recall also that we consider X as a subset of $\Delta[X]$ by identifying an element x in X with the singleton $\{x\} \subset X$ (see point 62).

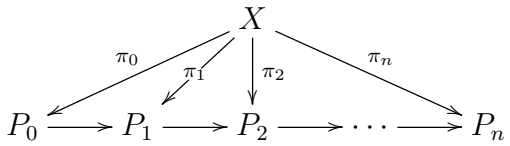
Let us choose $\psi\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$ an extension along $X \subset \Delta[X]$ (see point 62). By definition ψ maps a distance d on X to a distance on $\Delta[X]$ whose restriction along $X \subset \Delta[X]$ is d. For example ψ could be one of the extensions s, c, or a presented in Point 63. We are going to use the extension ψ to construct a dendrogram on X for any distance d on X. The construction follows the procedure explained in Point 109 and goes by induction.

Assume X is a finite and non-empty set, and d is a distance on X. By induction we are going to define the following table:

real numbers	$a_0 = 0$	a_1	$\cdots$	a_n
quotients of X	$\pi_0\colon X \to P_0$	$\pi_1\colon X \to P_1$	$\cdots$	$\pi_n\colon X \to P_n$
distances	(P_0, b_0)	(P_1, b_1)	$\cdots$	(P_n, b_n)

where

- the first row is a finite sequence of real numbers $0 = a_0 < a_1 < \cdots < a_n$;
- the second row is a finite sequence of quotients of X that fit into the following commutative diagram:



  and such that $|P_0| > |P_1| > \cdots > |P_n|$.
- the thrid row is a sequence of distances b_i on the corresponding sets P_i.

(0) Define:
  - $a_0 := 0$,
  - $P_0 := X/d_0$ and π_0 to be the quotient function $X \to X/d_0$ (see Point 25).
  - $b_0\colon P_0 \times P_0 \to [0, \infty]$ to map a pair of blocks (p, q) of the partition P_0, associated with the quotient, to the following element in $[0, \infty]$ (here we use the choice of an extension ψ):

$$\boxed{b_0(p, q) := \psi d(p, q)}$$

If b_0 is discrete, stop and set $n := 0$.

If b_0 is not discrete, consider its separation $\mathrm{sep}(b_0)$ and go to the next step.

(1) Define:
- $a_1 := \mathrm{sep}(b_0)$ (since b_0 is not discrete, $a_1 > 0$).
- d_1 to be the composition:

$$X \times X \xrightarrow{\pi_0 \times \pi_0} P_0 \times P_0 \xrightarrow{(b_0)_{\mathrm{sep}}} [0, \infty]$$

where $(b_0)_{\mathrm{sep}}$ is the discrete distance such that (see Point 25):

$$(b_0)_{\mathrm{sep}}(p, q) = \begin{cases} 0 & \text{if } b_0(p, q) = \psi d(p, q) \leq \mathrm{sep}(b_0) \\ \infty & \text{otherwise} \end{cases}$$

Note that if $d_0(x, y) = 0$, then $[x] = [y]$ in $P_0 = X/d_0$, and therefore $d_1(x, y) = 0$. It follows that $d_0 \geq d_1$.
- $P_1 := X/(d_1)$ and π_1 to be the quotient function $X \to X/(d_1)$. Do not confuse X/d_1 defined in Point 25 with $X/(d_1)$. Since $|X/(d_1)| = |P_0/(b_0)_{\mathrm{sep}}|$ and $|P_0| > |P_0/(b_0)_{\mathrm{sep}}|$ we have $|P_0| > |P_1|$.
- $b_1 \colon P_1 \times P_1 \to [0, \infty]$ to be a function such that:

$$\boxed{b_1(p, q) := \psi d(p, q)}$$

By definition b_1 is the restriction of ψd to the subset $P_1 = X/(d_1) \subset \Delta[X]$ and hence it is a distance.

If b_1 is discrete, stop and set $n := 1$.

If b_1 is not discrete, consider its separation $\mathrm{sep}(b_1)$ and go to the next step.

(2) Define:
- $a_2 := \mathrm{sep}(b_1)$,
- d_2 to be the composition:

$$X \times X \xrightarrow{\pi_1 \times \pi_1} P_1 \times P_1 \xrightarrow{(b_1)_{\mathrm{sep}}} [0, \infty]$$

Note that if $d_1(x, y) = 0$, then $[x] = [y]$ in $P_1 = X/(d_1)$, and therefore $d_2(x, y) = 0$. It follows that $d_1 \geq d_2$.
- $P_2 := X/(d_2)$ and π_2 to be the quotient function $X \to X/(d_2)$. Since $|X/d_2| = |P_1/(b_1)_{\mathrm{sep}}|$ and $|P_1| > |P_1/(b_1)_{\mathrm{sep}}|$ we have $|P_1| > |P_2|$.
- $b_2 \colon P_2 \times P_2 \to [0, \infty]$ to be a function such that:

$$\boxed{b_2(p, q) := \psi d(p, q)}$$

By definition b_2 is the restriction of ψd to the subset $P_2 = X/(d_2) \subset P(X)$ and hence it is a distance.

If b_2 is discrete, stop and set $n := 2$.

If b_2 is not discrete, consider its separation $\mathrm{sep}(b_2)$ and go to the next step.

(⋮) repeat the above process.

- Continue the above steps until $|P_n| = 1$, i.e., the partition P_n of X is trivial (see Point 4). Since the sizes of the obtained partitions form strictly decreasing sequence of numbers $|P_0| > |P_1| > \cdots > |P_n|$, this procedure indeed ends.

The first two rows of the obtained table define a dendrogram as explained in 109. We call the obtained dendrogram ψ **linkage** of the distance d and denoted it also by the symbol ψd.

118. In the previous point we described how an extension $\psi\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$ along $X \subset \Delta[X]$ leads to a function that maps a distance d on X to a dendrogram ψd on X. This function is called **hierarchical clustering** with respect to ψ. It is denoted also by the same symbol ψ:

$$\psi\colon \mathcal{D}(X) \to \mathrm{Dend}(X)$$

Recall that dendrograms on X are identified with ultra pseudometrics on X (see Point 112) via the functions:

$$\mathrm{Ultra}(X) \underset{u}{\overset{\mathcal{S}}{\rightleftarrows}} \mathrm{Dend}(X)$$

By composing ψ with u, to get a function:

$$\mathcal{D}(X) \xrightarrow{\ \psi\ } \mathrm{Dend}(X) \xrightarrow{\ u\ } \mathrm{Ultra}(X) \hookrightarrow \mathcal{D}(X)$$

In this way, an arbitrary distance on X is transformed into an ultra pseudometric. We think about this composition as self operator on the space of all distances $\mathcal{D}(X)$ on X. It is a simplifying process converting a distance into an ultra pseudometric that can be visualised via a dendrogram for example.

119. In Point 63 we defined 3 extensions along $X \subset \Delta[X]$:

$$\mathcal{D}(X) \underset{a}{\overset{s}{\underset{c}{\rightrightarrows}}} \mathcal{D}(\Delta[X])$$

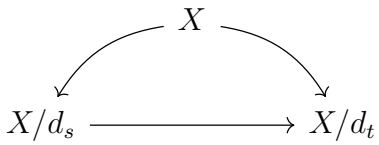
The hierarchical clusterings with resect to these extensions satisfy additional property:

120. **Proposition.** *Let $\psi\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$ be one of the extensions s or c or a above. Then the following diagram commutes:*

$$\mathrm{Ultra}(X) \hookrightarrow \mathcal{D}(X) \xrightarrow{\ \psi\ } \mathrm{Dend}(X), \qquad \mathrm{Ultra}(X) \xrightarrow{\ \mathcal{S}\ } \mathrm{Dend}(X)$$

121. **Exercise.** Prove Proposition 120.

122**. Exercise.** Consider the extension $s\colon \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$ and the induced hierarchical clustering $s\colon \mathcal{D}(X) \to \mathrm{Dend}(X)$. Show that for any distance space (X, d), the dendrogram sd is given by the following quotient functions which were defined in Point 25, for any $s < t$ in $[0, \infty)$:



Linear Algebra

123. **Vector spaces.** Let K be a field. We typically choose K to be the field with p-elements $\mathbf{F}_p$ for some prime p. For example, $\mathbf{F}_2$ consists of two elements $\{0, 1\}$ with the following operations:

$$0 + 0 = 0, 1 + 0 = 1, 1 + 1 = 0, 1 \cdot 0 = 0, 1 \cdot 1 = 1$$

A K **vector space** is a set V endowed with the following operations:

- addition: $V \times V \ni (v, w) \mapsto v + w \in V$ which is:
  - commutative $v + w = w + v$,
  - associative $(v + u) + w = v + (u + w)$
- a zero element $0 \in V$ such that $v + 0 = v$ for any v.
- multiplication by elements of the field: $K \times V \ni (a, v) \mapsto av \in V$ which is:
  - associative $a(bv) = (ab)v$,
  - distributive $a(v + w) = av + aw$ and $(a + b)v = av + bv$,
  - with 1 acting as the identity $1v = v$.

Note that by distributivity $v + (-1)v = (1 - 1)v = 0v = 0$. Thus any element v in V admits an additive inverse $-v$. The set V with the 0 element and addition is therefore an abelian group.

For example, for $n \geq 1$, the set K^n with the following operations is a K vector space:

$$(a_1, \ldots, a_n) + (b_1, \ldots, b_n) = (a_1 + b_1, \ldots, a_n + b_n) \qquad 0 = (0, \ldots, 0)$$

$$b(a_1, \ldots, a_n) = (ba_1, \ldots, ba_n)$$

We use the symbol 0 or K^0 to denote the unique vector space structure on the set with one element $\{0\}$.

124. **Basis.** Let V be a K vector space. Its elements are called **vectors**. A sequence $v_1, \ldots, v_n$ of vectors in V is called a **basis** if:

- the vectors $v_1, \ldots, v_n$ are linearly independent, i.e., equality $a_1v_1 + \cdots + a_nv_n = 0$ happens if and only if $a_1 = \cdots = a_n = 0$.
- the vectors in this sequence generate V, i.e., all elements of V can be written as linear combinations $a_1v_1 + \cdots + a_nv_n$.

For example, the following vectors form a basis of K^n called the **standard basis**:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \ldots e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

If $\beta = \{v_1, \ldots, v_n\}$ is a basis of V, then any v in V can be written in a unique way as a linear combination:

$$v = a_1 v_1 + \cdots + a_n v_n$$

The elements a_i are called the coordinates of v with respect to the basis β and we use the following notation:

$$_\beta[v] = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

The key property of vector spaces over a field is that all of them have bases.

A K vector space is called **finite dimensional** if it has a finite basis. If V is finite dimensional, then all its bases have the same number of elements. This number is called the **dimension** of the vector space and is denoted by $\dim(V)$.

125**.** In the case of the field with two elements $\mathbf{F}_2$, a vector space can be described more efficiently. An $\mathbf{F}_2$ vector space is a set V with a zero element $0 \in V$ and an addition operation $V \times V \ni (v, w) \mapsto v + w \in V$ such that, for any u, v, w in V:

- $v + w = w + v$ (commutativity),
- $(v + u) + u = v + (u + w)$ (associativity),
- $0 + v = v$
- $v + v = 0$

126**. Subspaces.** Let V be a K vector space. A subset $W \subset V$ is called a **subspace** if $0 \in W$ and, for any $v, w \in W$ and $a \in K$, $v + w \in W$ and $av \in W$. Thus a subspace of a K vector space is itself a K vector space. If V is finite dimensional, then so is any of its subspaces W and $\dim(W) \leq \dim(V)$.

127**. Quotients.** Let $W \subset V$ be a vector subspace. We say that two elements v and w in V are **equivalent modulo** W if $v - w \in W$. This is an equivalence relation. The equivalence classes of this relation are given by subsets of the form $[v] := v + W \subset V$. The symbol V/W denotes the set of equivalence classes of the equivalent modulo W relation. The set V/W with the following operations is a K vector space which is called the **quotient** of V by W:

- $(v + W) + (w + W) = v + w + W$
- $0 = 0 + W = W$
- $a(v + W) = av + W$

128**. Linear functions.** Let V and W be K vector spaces. A function $f \colon V \to W$ is called **linear** if for all v, w in V, and α in K:

$$f(0) = 0, \quad f(v + w) = f(v) + f(w), \quad f(\alpha v) = \alpha f(v)$$

For example, if $W \subset V$ is a subspace, then the inclusion $W \subset V$ and the quotient function $V \to V/W$ that maps v to $[v] = v + W$ are linear.

Note that if $f, g\colon V \to W$ are linear functions, then so are $f+g$, $f-g$ and af for any a in K.

A linear function $f\colon V \to W$ is called an **isomorphism** if and only if there is a linear function $g\colon W \to V$ such that $fg = \mathrm{id}$ and $gf = \mathrm{id}$. Vector spaces V and W are called **isomorphic** if and only if there is an isomorphism $V \to W$. Two finite dimensional vector spaces are isomorphic if and only if they have the same dimension. Thus any n-dimensional K vector space V is isomorphic to K^n. To produce an isomorphism between n-dimensional K vector space V and K^n we can choose a basis $\beta = \{v_1, \ldots, v_n\}$, the function $V \to K^n$ that maps a vector v to its coordinates $[v]_\beta$ with respect to the basis β is a linear isomorphism.

129**. Linear functions and matrices.** Let V be a finite dimensional vector space. Choose a basis $\beta = \{v_1, \ldots, v_n\}$ of V. For any choice of vectors $w_1, \ldots, w_n$ in a vector space W, there is a unique linear function $f\colon V \to W$ such that $f(v_i) = w_i$. For example, a linear function $f\colon K^n \to K^m$ is uniquely determined by n vectors $w_1, \ldots, w_n$ in K^m for which $f(e_i) = w_i$. In this way we can identify $f\colon K^n \to K^m$ with the $m \times n$ matrix of elements in K given by $[f] := \begin{bmatrix} w_1 & \cdots & w_n \end{bmatrix}$. This is the unique matrix for which $[f]v = f(v)$ where on the left a matrix multiplication in performed.

More generally, let V and W be finite dimensional vector spaces. Choose a basis $\beta = \{v_1, \ldots, v_n\}$ in V and a basis $\Gamma = \{w_1, \ldots, w_m\}$ in W. For any linear function $f\colon V \to W$, the matrix:

$$ {}_\Gamma[f]_\beta = \begin{bmatrix} {}_\Gamma[f(v_1)] & \cdots & {}_\Gamma[f(v_n)] \end{bmatrix} $$

is called the **matrix of f with respect to the basis β and Γ**. It is the unique $m \times n$ matrix of elements in K such that (where on the left a matrix multiplication is performed):

$$ {}_\Gamma[f]_\beta \; {}_\beta[v] = {}_\Gamma[f(v)] $$

Thus to represent a linear function $f\colon V \to W$ in form of a matrix, a convenient form for storing on a computer, three things need to be specified:

- a basis $\beta = \{v_1, \ldots, v_n\}$ in V
- a basis $\Gamma = \{w_1, \ldots, w_m\}$ in W
- $m \times n$ matrix M of elements in K.

If such a representation is given, then to retrieve the value $f(v)$ we need to:

- find the coordinates of v in basis β:

$$ v = a_1 v_1 + \cdots + a_n v_n $$

- take matrix multiplication:

$$ M \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} $$

- take the linear combination $f(v) = b_1 w_1 + \cdots + b_m w_m$.

130. **Kernels, images and cokernels.** Let $f\colon V \to W$ be a linear function between vector spaces. Then:

- $\mathrm{Ker}(f) = \{v \in V \mid f(v) = 0\} \subset V$ is a subspace called the **kernel** of f.
- $\mathrm{im}(f) = \{f(v) \mid v \in V\} \subset W$ is a subspace called the **image** of f.
- $\mathrm{coker}(f) = W/\mathrm{im}(f)$ is a vector space called the **cokernel** of f.

A linear function $f\colon V \to W$ leads to several new linear functions:

$$(*) \qquad \mathrm{Ker}(f) \hookrightarrow V \twoheadrightarrow \mathrm{im}(f) \hookrightarrow W \twoheadrightarrow \mathrm{coker}(f), \quad V \xrightarrow{f} W$$

where the functions indicated by the arrows $\twoheadrightarrow$ are surjective and by the arrows $\hookrightarrow$ injective.

Recall that if V and W are finite dimensional, then:

$$\dim(V) = \dim(\ker(f)) + \dim(\mathrm{im}(f)) \qquad \dim(W) = \dim(\mathrm{im}(f)) + \dim(\mathrm{coker}(f))$$

131. **Exercise.** Let a linear function $f\colon V \to W$ be represented by:

- a basis $\beta = \{v_1, \ldots, v_n\}$ in V
- a basis $\Gamma = \{w_1, \ldots, w_m\}$ in W
- an $m \times n$ M matrix of elements in K.

Find such representations for all the functions in the diagram $(*)$ above. This is not an easy exercise. You need to use Gaussian elimination.

132. **Proposition.** *Consider the following commutative square of linear functions between vector spaces:*

$$\begin{array}{ccc} V_1 & \xrightarrow{f} & V_0 \\ {\scriptstyle \alpha_1}\downarrow & & \downarrow{\scriptstyle \alpha_0} \\ W_1 & \xrightarrow{g} & W_0 \end{array}$$

Then there are unique linear functions $k\colon \ker(f) \to \ker(g)$, $i\colon \mathrm{im}(f) \to \mathrm{im}(g)$, *and* $c\colon \mathrm{coker}(f) \to \mathrm{coker}(g)$ *making the following diagram commutative:*

$$\begin{array}{ccccccccc} \ker(f) & \hookrightarrow & V_1 & \twoheadrightarrow & \mathrm{im}(f) & \hookrightarrow & V_0 & \twoheadrightarrow & \mathrm{coker}(f) \\ {\scriptstyle k}\downarrow & & {\scriptstyle \alpha_1}\downarrow & & \downarrow{\scriptstyle i} & & {\scriptstyle \alpha_0}\downarrow & & \downarrow{\scriptstyle c} \\ \ker(g) & \hookrightarrow & W_1 & \twoheadrightarrow & \mathrm{im}(g) & \hookrightarrow & W_0 & \twoheadrightarrow & \mathrm{coker}(g) \end{array}, \quad V_1 \xrightarrow{f} V_0, \quad W_1 \xrightarrow{g} W_0$$

133. **Exercise.** Prove Proposition 132.

134. **Sets and vector spaces.** Choose a field K. For any finite set X, define $KX := \oplus_{x \in X} K$. The vector space KX is isomorphic to $K^{|X|}$.

The elements of $KX = \oplus_{x \in X} K$ are sequences $(v_x)_{x \in X}$ indexed by elements of X. For example, the following vectors form what we call the standard basis of KX. Let x be in X. Define e_x to be an element in KX given by the sequence:

$$(e_x)_y = \begin{cases} 1 & \text{if } y = x \\ 0 & \text{if } y \neq x \end{cases}$$

Thus KX has a basis indexed by elements of X, and hence it is of dimension $|X|$. We often identify an element x in X with the vector e_x in KX. With that identification, any element in KX can be written uniquely as a linear combination $\sum_{x \in X} a_x x$.

Let $f \colon X \to Y$ be a map of sets. Define $Kf \colon KX \to KY$ to be the unique linear function such that $(Kf)(x) = f(x)$. An element $\sum_{x \in X} a_x x$ in KX is mapped via Kf to $\sum_{x \in X} a_x f(x)$ in KY. If we choose X and Y as basis of respectively KX and KY, then Kf is represented by a $|Y| \times |X]$ matrix $M = [m_{yx}]_{y \in Y, x \in X}$ where:

$$m_{yx} = \begin{cases} 1 & \text{if } y = f(x) \\ 0 & \text{if } y \neq f(x) \end{cases}$$

Thus all the columns of such a matrix have all entries equal to 0 except for one entry whose value is 1.

Note that $K\mathrm{id} = \mathrm{id}$ and $K(fg) = K(f)K(g)$.

Vector spaces parametrised by $[0, \infty)$.

Throughout the rest of these notes, let us choose and fix a field K. All considered vector spaces are vector spaces over K.

135. Similarly to a tame set parametrised by $[0, \infty)$ (see Point 101), a **tame vector space parametrised** by $[0, \infty)$ is sequence of finite dimensional vector spaces V_t, indexed by t in $[0, \infty)$, together with a sequence of linear functions $V_{s<t} \colon V_s \to V_t$, called **transitions**, indexed by pairs $s < t$ of elements in $[0, \infty)$. These functions are required to satisfy the following properties:

(1) For any $r < s < t$ in $[0, \infty)$ the following diagram commutes:

$$\begin{array}{ccccc} V_r & \xrightarrow{V_{r<s}} & V_s & \xrightarrow{V_{s<t}} & V_t \\ & \searrow & & \nearrow & \\ & & V_{r<t} & & \end{array}$$

(2) There is a sequence of real numbers $0 = a_0 < a_1 < \cdots < a_n$ such that the transition $V_{s<t} \colon V_s \to V_t$ is an isomorphism for all pairs $s < t$ for which both s and t belong to the same interval among $[a_0, a_1), \ldots, [a_{n-1}, a_n), [a_n, \infty)$. We also say that the sequence $0 = a_0 < a_1 < \cdots < a_n$ **discretises** V.

The symbol $\mathrm{tame}([0, \infty), \mathrm{Vect})$ denotes the set of tame vector spaces parametrised by $[0, \infty)$.

Let V be a vector space. Then the same symbol V denotes also the **constant** tame parametrised vector space such that $V_t = V$ and the transition $V_{s<t} \colon V \to V$ is the identity for all $s < t$ in $[0, \infty)$.

136. **Exercise.** Let V be a tame vector space parametrised by $[0, \infty)$ and $0 = a_0 < a_1 < \cdots < a_n$ be a sequence that discretises V. Show that by adding any elements to this sequence we obtain a sequence that also discretises V. Conclude that any finite collection of tame vector spaces parametrised by $[0, \infty)$ have a common sequence that discretises all of them.

137. **Maps.** Let V and W be tame vector spaces parametrised by $[0, \infty)$. A **map** between V and W, denoted by $f \colon V \to W$, is a sequence of linear functions $f_t \colon V_t \to W_t$ for which the following square commutes for all $0 \leq s < t$ in $[0, \infty)$:

$$\begin{array}{ccc} V_s & \xrightarrow{V_{s<t}} & V_t \\ {\scriptstyle f_s}\downarrow & & \downarrow{\scriptstyle f_t} \\ W_s & \xrightarrow{W_{s<t}} & W_t \end{array}$$

Such a map f is:

(1) an **isomorphism** if f_t is an isomorphism for all t;
(2) an **epimorphism** if f_t is an epimorphism for all t;
(3) a **monomorphism** if f_t is an monomorphism for all t.

Two parametrised vector spaces V and W are said to be **isomorphic**, if there is an isomorphism between them.

Note that if $f, g\colon V \to W$ are maps, then so are $f+g$, $f-g$ and af for any a in K. With these operations, the set of maps between V and W $\mathrm{map}(V, W)$ becomes a vector space.

138**. Exercise.** Let V and W be vector spaces. Show that the set of all maps between the constant vector spaces V and W parametrised by $[0,\infty)$ is in bijection with the set of linear functions between the vector spaces V and W.

139**. Exercise.** Show that a tame vector space X parametrised by $[0,\infty)$ is isomorphic to a constant vector space parametrised by $[0,\infty)$ if and only if $X_{s<t}\colon X_s \to X_t$ is an isomorphism for all $s<t$ in $[0,\infty)$.

140**.** To define a tame vector space parametrised by $[0,\infty)$ we can use a table:

real numbers	$0=a_0$	a_1	a_2	$\cdots$
matrices of elements in the field K	M_0	M_1	M_2	$\cdots$

where

- the first row is a finite sequence of real numbers $0=a_0<a_1<\cdots<a_n$;
- the second row is a finite sequence $M_0,\dots M_{n-1}, M_n = I$ of matrices of elements in K (the last one being the identity matrix) such that the numbers of rows in M_i is equal to the numbers of columns in M_{i+1} so that it makes sense to take the matrix multiplication $M_{i+1}M_i$.

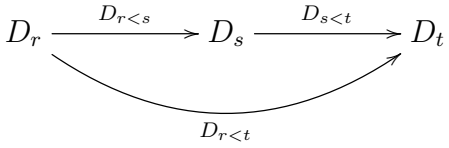
Here is how such a table is used to obtain a tame vector space parametrised by $[0,\infty)$:

- For $t\in[a_i, a_i+1)$ let $n_t =$ (number of columns of M_i) and define $V_t := K^{n_t}$.
- for $0\le s<t$, with $s\in[a_i,a_{i+1})$ and $t\in[a_j,a_{j+1})$ set the linear function $W_{s<t}\colon W_s = K^{n_s}\to K^{n_t}=W_t$ to be given with respect to the standard basis by the matrix multiplication $M_{j-1}\cdots M_i$.

Note that the sequence $0=a_0<a_1<\cdots<a_n$ discretises the constructed tame parametrised vector space.

141**. Exercise.** Prove that any vector space parametrised by $[0,\infty)$ is isomorphic to one that is described by the procedure in Point 140.

142**. From parameterised sets to parametrised vector spaces.** Consider a tame set D parametrised by $[0,\infty)$ represented by functions for all $r<s<t$ in $[0,\infty)$:



By taking the K vector spaces generated by all the sets involved in D together with linear functions induced by all the functions of sets in D we can transform such a parametrised set into a vector space parametrised by $[0,\infty)$ (see Point 134). We denote it by KD. It is given by the linear functions:

$$KD_r \xrightarrow{KD_{r<s}} KD_s \xrightarrow{KD_{s<t}} KD_t \qquad KD_r \xrightarrow{KD_{r<t}} KD_t$$

. If $0 = a_0 < a_1 < \cdots < a_n$ discretises D, then this sequence discretises also KD and hence KD is indeed a tame vector space parametrised by $[0,\infty)$.

143. **Bars.** Let $0 \leq a < b \leq \infty$ be elements in $[0,\infty)$. For any t in $[0,\infty)$, define:

$$[a,b)_t = \begin{cases} K & \text{if } a \leq t < b \\ 0 & \text{otherwise .} \end{cases}$$

For every $s < t$ in $[0,\infty)$, set $[a,b)_{s<t} \colon [a,b)_s \to [a,b)_t$ to be either 0, if $[a,b)_s$ or $[a,b)_t$ is the zero vector space, or the identity function if $[a,b)_s = [a,b)_t = K$. These vector spaces and functions form a tame vector space parametrised by $[0,\infty)$ which we denote by $[a,b)$ and call the **bar starting at** a **and ending at** b. If $a < b < \infty$, then the sequence $0 < a < b$ discretises $[a,b)$. The sequence $0 < a$ discretises $[a,\infty)$.

144. **Maps out of** $[a,\infty)$**.** Assume $a \in [0,\infty)$. Let V be a tame vector space parametrised by $[0,\infty)$. Choose an element v in V_a and consider the induced linear function $v_a \colon K \to V_a$ that maps 1 to v. For $t > a$ define $v_t \colon K \to V_t$ to be the composition:

$$K \xrightarrow{v_a} V_a \xrightarrow{V_{a<t}} V_t$$

Then these linear functions form a map denoted by the same symbol $v \colon [a,\infty) \to V$.

Vice versa, if $f \colon [a,\infty) \to V$ is a map, then it is induced by the element $f_a(1)$ in V_a. Furthermore, the function that sends a map $f \colon [a,\infty) \to V$ to $f_a(1)$ in V_a is a linear isomorphism between the vector spaces $\mathrm{map}([a,\infty), V)$ and V_a. In particular:

$$\mathrm{map}([a,\infty),[c,d)) = \begin{cases} \text{is 1 dimensional} & \text{if } c \leq a < d \\ 0 & \text{otherwise .} \end{cases}$$

145. **Maps out of** $[a,b)$**.** Assume $a < b < \infty$. Consider a map $f \colon [a,b) \to V$. Let $[a,\infty) \to [a,b)$ be the map induced by the identity $\mathrm{id} \colon [a,\infty)_a = K \to K = [a,b)_a$. Since this map is a surjection, precomposing with it induces an inclusion:

$$\mathrm{map}([a,b),V) \subset \mathrm{map}([a,\infty),V) = V_a$$

. The image of this inclusion coincides with the kernel $\ker(V_{a<b} \colon V_a \to V_b)$ (see Exercise 146). For an element v in $\ker(V_{a<b} \colon V_a \to V_b)$ we then use the symbol $v \colon [a,b) \to V$ to denote the map such that $v_a(1) = v$.

In particular:

$$\operatorname{map}([a,b),[c,d)) = \begin{cases} \text{is 1 dimensional} & \text{if } c \le a < d \le b \\ 0 & \text{otherwise .} \end{cases}$$

146. **Exercise.** Show that the image of $\operatorname{map}([a,b),V) \subset \operatorname{map}([a,\infty),V) = V_a$ coincides with the kernel $\ker(V_{a<b}\colon V_a \to V_b)$.

147. **Exercise.** Explain the equality describing the dimension of $\operatorname{map}([a,b),[c,d))$ in Point 145.

148. **Direct sums.** Let V and W be tame vector spaces parametrised by $[0,\infty)$ given by the linear functions $\{V_{s<t}\colon V_s \to V_t\}$ and $\{W_{s<t}\colon W_s \to W_t\}$. Define $V \oplus W$ to be the tame vector space parametrised by $[0,\infty)$ given by the following functions for every $s < t$ in $[0,\infty)$:

$$V_{s<t} \oplus W_{s<t}\colon V_s \oplus W_s \to V_t \oplus W_t$$

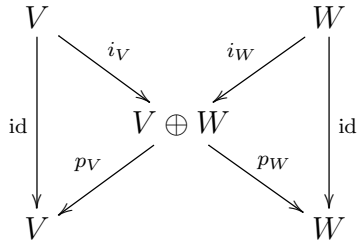
The parametrised vector space $V \oplus W$ is called the **direct sum of V and W**. The inclusions $i_V\colon V \to V \oplus W$ and $i_W\colon W \to V \oplus W$:

$$(i_V)_t\colon V_t \subset V_t \oplus W_t, v \mapsto (v,0) \qquad (i_W)_t\colon W_t \subset V_t \oplus W_t, w \mapsto (0,w)$$

and the projections $p_V\colon V \oplus W \to V$ and $p_W\colon V \oplus W \to W$:

$$(p_V)_t\colon V_t \oplus W_t \to V_t, (v,w) \mapsto v \qquad (p_W)_t\colon V_t \oplus W_t \to W_t, (v,w) \mapsto w$$

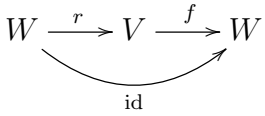
are maps that fit into the following commutative diagram:



149. **Indecomposables.** A tame vector space parametrised by $[0,\infty)$ is called **indecomposable** if it is not isomorphic to a direct sum of two non-zero tame vector spaces parametrised by $[0,\infty)$.

150. **Exercise.** Show that for every $a < b$ in $[0,\infty]$, the bar $[a,b)$ is indecomposable.

151. **Exercise.** How can one show that a tame vector space V parametrised by $[0,\infty)$ is isomorphic to a direct sum $W \oplus U$? Assume that there is a commutative diagram of transformations of the form:



If such a diagram is given, prove that the transformation is an isomorphism:

$$\begin{bmatrix} r \\ \text{inclusion} \end{bmatrix} : W \oplus \ker(f) \to V$$

Rank and bar decomposition.

152. **Rank.** Let V be a tame vector space parametrised by $[0,\infty)$. Choose a sequence $0 = a_0 < a_1 < \cdots < a_n$ that discretises V (see Point 135) and consider the following natural number:

$$\dim V_0 + \dim \operatorname{coker}(V_{0<a_1}) + \dim \operatorname{coker}(V_{a_1<a_2}) + \cdots + \dim \operatorname{coker}(V_{a_{n-1}<a_n})$$

We claim that this sum does not depend on the choice of the sequence $a_0 < \cdots < a_n$. To see this note that if we modify the sequence by adding an extra element say $b \in (a_i, a_{i+1})$, then since $\operatorname{coker}(V_{a_i<b}) = 0$ and $\operatorname{coker}(V_{b<a_{i+1}}) = \operatorname{coker}(V_{a_i<a_{i+1}})$, the sums for both of the sequences $0 = a_0 < \cdots < a_n$ and $0 = a_0 < \cdots < a_i < b < a_{i+1} < \cdots < a_n$ are the same.

By adding elements we can refine two sequences that discretise V to a common sequence. As each such addition does not modify the sums our claim stands.

The number given by the above sum is called the **rank** of V and is denoted by $\operatorname{rank}(V)$. For example $\operatorname{rank}([b,e)) = 1$.

153. **Exercise.** Prove that $\operatorname{rank}(V \oplus W) = \operatorname{rank}(V) + \operatorname{rank}(W)$ for all vector spaces V and W parametrised by $[0,\infty)$. Conclude that $\operatorname{rank}(\oplus_{i=1}^{r}[a_i,b_i)) = r$.

154. **Exercise.** Assume $f\colon V \to W$ is an epimorphism. Show that $\operatorname{rank}(V) \geq \operatorname{rank}(W)$.

155. **Generating tame parametrised vector spaces.** Let V be a tame vector space parametrised by $[0,\infty)$. Consider a sequence of elements $\{v_i \in V_{a_i}\}_{i\in I}$ and the induced map (see Point 144):

$$\sum_{i\in I} v_i\colon \ \oplus_{i\in I}\,[a_i,\infty) \to V$$

If this map is an epimorphism, then the sequence $\{v_i \in V_{a_i}\}_{i\in I}$ is said to **generate** V. V is called **finitely generated** if there is a finite sequence generating V. For example any non zero element in $K = [a,b)_a$ generates $[a,b)$.

156. **Exercise.** Prove that a tame vector space V parametrised by $[0,\infty)$ is generated by one element if and only if V is isomorphic to a bar $[a,b)$ for some $a < b$ in $[0,\infty]$.

157. **Proposition.** *Let V be a tame vector space parametrised by $[0,\infty)$. Then $\operatorname{rank}(V)$ is equal to the minimal number of elements that generates V. In particular, V is finitely generated.*

Proof. Let $0 = a_0 < a_1 < \cdots < a_n$ be sequence that discretises to V. Assume $\operatorname{rank}(V) = n$. Let $v_{i,j}$ in V_{a_i} be vectors such that:

- $\{v_{0,j}\}_{1\leq j\leq J_0}$ is a base in V_0

- the images of $\{v_{ij}\}_{1\leq j\leq J_i}$ form a base of $\operatorname{coker}(V_{a_{i-1}<a_i}\colon V_{a_{i-1}} \to V_{a_i})$ for every $1 \leq i \leq n$.

Then the following induced map is an epimorphism:

$$\sum_{i,j} v_{i,j}\colon \ \oplus_{i,j}\,[a_i,\infty) \to V$$

The minimal number of generators of V can not therefore be bigger than its rank.

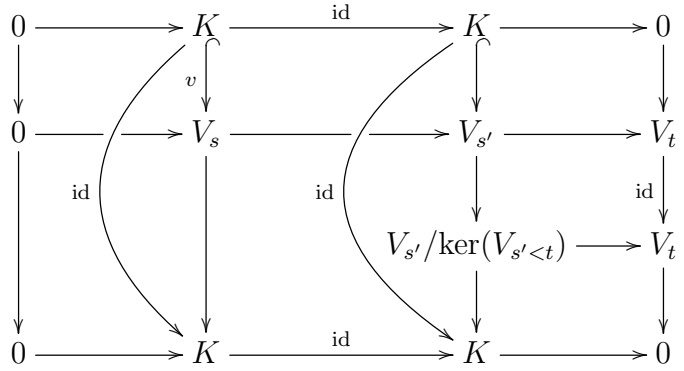
Let $\oplus_{k=1}^{m} K(s_k,\infty) = F \to V$ be an epimorphism. Choose a sequence $0 = a_0 < a_1 < \cdots < a_n$ that discretises both F and V. For any i, since $F_{a_i} \to V_{a_i}$ is surjective, then so are $F_0 \to V_0$ and $\operatorname{coker}(F_{a_i<a_{i+1}}) \to \operatorname{coker}(V_{a_i<a_{i+1}})$. It follows that $m = \operatorname{rank}(F) \geq \operatorname{rank}(V)$. Thus the rank of V can not be bigger than its minimal number of generators. □

We are now ready to state and prove the key decomposition result for tame vector spaces parametrised by $[0,\infty)$.

158**. Theorem.** *Any tame vector space parametrised by* $[0,\infty)$ *is isomorphic to a direct sum* $\oplus_{i=1}^{r}[a_i,b_i)$.

Such a decomposition into a direct sum of bars is called a **bar decomposition**.

Proof. Assume the theorem is not true. Let V be a contra example to the theorem with the smallest rank. Let s be the smallest number such that $V_s \neq 0$. Choose a non zero element v in V_s. Let t be the minimal element such that $V_{s<t}(v) = 0$. For any $s < s' < t$ we can build the following commutative diagram:



This shows (see Exercise 151) that V is isomorphic to $[s,t) \oplus (V/[s,t))$. Thus since $\operatorname{rank}(V/[s,t)) = \operatorname{rank}(V) - 1$ is smaller than $\operatorname{rank}(V)$ according to our assumption $\operatorname{rank}(V/[s,t))$ is a direct sum of bars. It follows that so is V, which contradicts our assumption about the existence of a contra example to the theorem. The theorem is therefore true. □

159**. Corollary.** *Let* V *be a tame vector space parametrised by* $[0,\infty)$. *Then* $\operatorname{rank}(V)$ *is the number of bars in any bar decomposition of* V. *Consequently any bar decomposition of* V *has the same number of bars.*

160. **Theorem.** *Assume* $\oplus_{i=1}^{n}[a_i, b_i)$ *and* $\oplus_{j=1}^{m}[c_j, d_j)$ *are isomorphic vector spaces parametrised by* $[0, \infty)$. *Then:*

- $n = m$
- *There is a permutation* α *of the set* $\{1, \ldots, n\}$ *so that* $a_i = c_{\alpha(i)}$ *and* $b_i = d_{\alpha(i)}$ *for any* i.

Proof. The fact that $n = m$ follows from Corollary 159. The second statement we prove by induction on n. If $n = 1$ the theorem is clear. Assume that the theorem is true for any $n < k$.

Choose an isomorphism $f\colon \oplus_{i=1}^{n}[a_i, b_i) \to \oplus_{j=1}^{m}[c_j, d_j)$. Let a_1 (possibly after taking a permutation) be the smallest among all a_i. After again possible permutation, let b_1 be the largest among the ends of the bars that start at a_1. We can assume the same about c_1 and d_1. Furthermore, we can choose c_1 and d_1 so that the following map $[a_1, b_1) \to [c_1, d_1)$ is an isomorphism:

$$[a_1, b_1) \hookrightarrow \oplus_{i=1}^{n}[a_i, b_i) \xrightarrow{f} \oplus_{j=1}^{m}[c_j, d_j) \xrightarrow{\text{pr}} [c_1, d_1)$$

If follows that $\oplus_{i=2}^{n}[a_i, b_i)$ and $\oplus_{j=2}^{m}[c_j, d_j)$ are isomorphic. By induction, we can then conclude that the theorem is true. □

161. **Exercise.** Let V and W be finitely generated vector spaces parametrised by $[0, \infty)$. Assume there is a monomorphism $f\colon W \subset V$ such that there is t for which $f_t\colon W_t \to V_t$ is not an isomorphism (a proper inclusion). Prove that $\text{rank}(W) < \text{rank}(V)$.

162. **Multi subsets.** Let T be a set. A **finite multi subset** of T is a function $f\colon T \to \mathbf{N}$ whose support:

$$\text{supp}(f) := \{t \in T \mid f(t) \neq 0\}$$

is a finite subset of T.

Let $f\colon T \to \mathbf{N}$ be a finite multi subset of T. Then:

- $f(t)$ is called the multiplicity of t.
- $\sum_{t \in T} f(t)$ is called the **size** or **cardinality** of f.

Let $f, g\colon T \to \mathbf{N}$ be finite multi subsets of T. Then f is called a **subset** of g if $f(t) \leq g(t)$ for any t in T.

163. **Multi subsets of upper quadrant of the extend plane.** Let V be a tame vector space parametrised by $[0, \infty)$. According to Theorems 158 and 160 there is a unique sequence of elements $a_1 < b_1, a_2 < b_2, \ldots, a_n < b_n$ in $[0, \infty]$ such that V is isomorphic to $\oplus_{i=1}^{n}[a_i, b_i)$.

Set:

$$T := \{(a, b) \in [0, \infty) \times [0, \infty] \mid a < b\}$$

This set is called the **upper quadrant of the extend plane**. Define $\beta_V : T \to \mathbf{N}$ to be the finite multi subset of T such that V is isomorphic to:

$$\oplus_{i=1}^{n}[a_i, b_i) = \oplus_{(a<b)\in T}[a, b)^{\beta_V(a<b)}$$

Theorems 158 and 160 can be then reformulated as:

164**. Corollary.** *The association $V \mapsto \beta_V$ is a bijection between the set of isomorphism classes of tame vector spaces parametrised by $[0, \infty)$ and finite multi subsets of the upper quadrant of the extend plane T.*

Pseudometrics on tame($[0,\infty)$, Vect).

In this section we explain how to endow the set tame($[0,\infty)$, Vect) with various metrics.

165. **Actions of** $([0,\infty),+,0)$ **on then poset** $[0,\infty)$**.** We are going to associate a metric on tame($[0,\infty)$, Vect) with every action of the additive monoid $([0,\infty),+,0)$ on the poset $[0,\infty)$. Such an action is a function $*\colon [0,\infty]\times[0,\infty)\to[0,\infty)$, $(x,t)\mapsto x*t$ such that:

- it preserves the order $y*s\geq x*t$ for all $y>x$ and $s>t$ in $[0,\infty)$;
- $x*0=x$ for all x in $[0,\infty)$;
- $(x*t)*s=x*(t+s)$.

Note that $x=x*0\leq x*t$, for any x and t in $[0,\infty)$.

166. **Exercise.** Show that $x*t=x+t$ is an action of the additive monoid $([0,\infty),+,0)$ on the poset $[0,\infty)$. It is called the **standard** action.

167. Let $f\colon[0,\infty)\to(0,\infty)$ be an integrable function with strictly positive values. Since f is integrable with strictly positive values, for any x and t in $[0,\infty)$, the set $\{y\mid x\leq y \text{ and } \int_x^y f(u)du=t\}$ contains only one element. Define $x*t$ to be the unique element in $[x,\infty)$ such that:

$$\int_x^{x*t} f(u)du=t$$

We claim that this defines an action of the additive monoid $([0,\infty),+,0)$ on the poset $[0,\infty)$. This is a consequence of additivity of integrals:

$$\int_x^{(x*t)*s} f(u)du=\int_x^{x*t} f(u)du+\int_{x*t}^{(x*t)*s} f(u)du=t+s=\int_x^{x*(t+s)} f(u)du$$

168. **Pseudometrics associated with actions.** Let $*$ be an action of $([0,\infty),+,0)$ on the poset $[0,\infty)$ and ϵ be in $[0,\infty)$.

- A map $f\colon V\to W$ between tame vector spaces parametrised by $[0,\infty)$ is called **ϵ-equivalence** , for every x in $[0,\infty)$, there is a linear function $s_x\colon W_x\to V_{x*\epsilon}$ making the following diagram commutative:

$$\begin{array}{ccc} V_x & \xrightarrow{V_{x<x*\epsilon}} & V_{x*\epsilon} \\ {\scriptstyle f_x}\downarrow & \overset{s_x}{\nearrow} & \downarrow{\scriptstyle f_{x*\epsilon}} \\ W_x & \xrightarrow[W_{x<x*\epsilon}]{} & W_{x*\epsilon} \end{array}$$

- Two tame vector spaces V and W parametrised by $[0,\infty)$ are called **ϵ-close** if:
  - there is τ_1 equivalence $U\to V$,

  - there τ_2 equivalence $U \to W$,
  - $\tau_1 + \tau_2 \leq \epsilon$.
- Let V and W be tame vector spaces parametrised by $[0, \infty)$.

$$d(V, W) := \inf\{\epsilon \mid \ V \text{ and } W \text{ are } \epsilon\text{-close}\}$$

  is called the distance induced by the action $*$.

169. **Theorem.** *Let* $*\colon [0, \infty] \times [0, \infty) \to [0, \infty)$ *be an action as defined in 165*

Simplicial complexes

Throughout the note we are going to choose and fix a set $\mathcal{U}$ called a universe.

170. A **simplicial complex** is a collection K of some finite non empty subsets of $\mathcal{U}$ that satisfies the following requirement: if $\sigma \subset \mathcal{U}$ is in K then any non-empty subset of σ is also in K.

Let $X \subset \mathcal{U}$ be a subset. The collection $\{\{x\} \mid x \in X\}$, consisting of singletons in X, is a simplicial complex denoted also by the symbol X and called the **discrete** simplicial complex on X. The collection $\{\sigma \subset X \mid 1 \leq |\sigma| < \infty\}$ of all finite non empty subsets of X is also a simplicial complex denoted by $\Delta[X]$ and called **the simplex on** X.

171. **Simplices.** Let K be a simplicial complex. An element $\sigma = \{x_0, \ldots, x_n\}$ in K is called a simplex of dimension $|\sigma| - 1 = n$. The set of n-dimensional simplices is denoted by K_n. An element $x \in \mathcal{U}$ is called a **vertex** of K if $\{x\}$ is a simplex in K. The assignment $x \mapsto \{x\}$ is a bijection between the set of vertices in K and its set of 0-dimensional simplices K_0. We use this bijection to identify these sets. Thus we are going to refer to 0-dimensional simplices also as vertices. Simplices of dimension 1 are called **edges**.

The maximum n such that K_n is non empty is called the **dimension of** K.

Let K be a simplicial complex. The collection $\cup_{0 \leq i \leq n} K_i$ is also a simplicial complex. It is a subcomplex of K called the n-th **skeleton** of K and is denoted by $\mathrm{sk}_n K$.

Let X be a subset of $\mathcal{U}$. We think about a simplicial complex K with $K_0 = X$ as a structure on X and we use the symbol $\mathrm{Simp}(X)$ to denote the collection of all simplicial complexes K such that $K_0 = X$.

172. **The lattice of simplicial complexes.** Note that if $\{K^i\}_{i \in I}$ is a family of simplicial complexes, then both the intersection $\cap_{i \in I} K^i$ and the union $\cup_{i \in I} K^i$ are also simplicial complexes. For example, if K is a simplicial complex and $X \subset \mathcal{U}$ is a subset, then the intersection $K \cap \Delta[X]$ is a simplicial complex consisting of these elements of K that are subsets of X. This intersection is called the **restriction** of K to X and is denoted by K_X.

Let L and K be simplicial complexes. If $L \subset K$, then L is called a **subcomplex** of K. Being a subcomplex is a partial order relation on the collection of all simplicial complexes which gives this collection the structure of a lattice. The union is the join and the intersection is the meet.

173. **Disjoint unions.** Two simplicial complexes L and K are said to be **disjoint** if $K \cap L = \emptyset$.

A simplicial complex K is a **disjoint union** of simplicial complexes $L_1, \ldots, L_k$ if $K = \cup_{i=1}^k L_i$ and L_i and L_j are disjoint for all pairs $i \neq j$. If K is a disjoint union of $L_1, \ldots, L_k$, then we write $K = \coprod_{i=1}^k L_i$.

Let X be a subset of $\mathcal{U}$ and $P = \{B_1, \ldots, B_k\}$ be a partition of X. We use the symbol $\Delta[P]$ to denote the disjoint union $\coprod_{i=1}^{k} \Delta[B_i]$. Note that $\Delta[P]$ is the subcomplex of $\Delta[X]$ consisting of these subsets of X that are included entirely in some block of P. For example the 0-th skeleton $\mathrm{sk}_0 K$ of a simplicial complex K coincide with $\Delta[S]$ where S is the singleton partition of K_0 (see 4).

Note that two partitions P and Q of X are equal if and only if the corresponding simplicial complexes $\Delta[P]$ and $\Delta[Q]$ are equal. Thus by mapping a partition P to the simplicial complex $\Delta[P]$ we obtain an inclusion:

$$\Delta \colon \mathrm{Part}(X) \subset \mathrm{Simp}(X)$$

In this way we can think about partitions of X as certain simplicial complexes on X.

174. **Connectivity.** Let K be a simplicial complex. Two vertices v and w in K are said to be in the **same component**, if there is a sequence of vertices $v = x_0, \ldots, x_n = w$ in K such that $\{x_i, x_{i+1}\}$ is an edge in K for every i. The symbol $[v] \subset K_0$ denotes the subset of vertices which are in the same component as v.

Being in the same component is an equivalence relation on K_0 whose equivalence classes are given by subsets of the form $[v]$. These subsets therefore form a partition of K_0 which is denoted by $\pi_0(K)$.

A simplicial complex K is called **connected** if $\pi_0(K)$ has only one element, i.e., if all vertices are in the same component. For example $\Delta[X]$ is connected for any non empty subset X in $\mathcal{U}$.

Let X be a subset of $\mathcal{U}$. By assigning to a simplicial complex K for which $K_0 = X$, the partition $\pi_0(K)$ of X, we obtain a function:

$$\pi_0 \colon \mathrm{Simp}(X) \to \mathrm{Part}(X)$$

If a simplicial compex is induced by a partition and is given by $\Delta[P]$, then we can recover the partition P by applying π_0 as we claim that the following diagram commutes:

$$\begin{array}{ccccc} \mathrm{Part}(X) & \xrightarrow{\Delta} & \mathrm{Simp}(X) & \xrightarrow{\pi_0} & \mathrm{Part}(X) \\ & \searrow & & \nearrow_{\mathrm{id}} & \end{array}$$

175. **Exercise.**

(1) Prove that $\pi_0(\Delta[P]) = P$ for any partition P of X, i.e., the above diagram commutes.
(2) Explain why $\Delta[P]$ is connected if and only if P is the trivial partition $\{X\}$.

176. **Connected components.** Let K be a simplical complex. Assume $\pi_0(K) = \{B_1, \ldots, B_k\}$ and consider sub-complexes of K given by the restrictions $K_{B_i} = K \cap \Delta[B_i]$ (see Point 172). These sub-complexes are disjoint and, since any simplex in K is a simplex in some of these sub-complexes, we get:

$$K = \coprod_{B \in \pi_0(K)} K_B$$

Note that for every i, the simplicial complex K_{B_i} is connected. That is the reason why these sub-complexes are called **connected components** of K. We also refer to $\pi_0(K)$ as to the **set of connected components of** K. For example if $P = \{B_1, \ldots, B_k\}$ is a partition of X, then $\Delta[B_i] \subset \Delta[P]$ is the connected component of $\Delta[P]$ corresponding to the element B_i in $\pi_0(\Delta[P]) = P$ (see Exercise 175).

Maps between simplicial complexes

177. A **map** between two simplicial complexes K and L is by definition a function $\phi\colon K \to L$ for which there exists a function $f\colon K_0 \to L_0$ such that $\phi(\sigma) = \cup_{x\in\sigma} f(\{x\})$ for any $\sigma \in K$. In particular $\phi(\{x\}) = f(\{x\})$ for any vertex x in K. Thus f is uniquely determined by ϕ and we often use the symbol ϕ_0 to denote f. Furthermore if K and L are fixed, then ϕ is determined by $f = \phi_0$, in which case we say that ϕ is induced by f.

The inclusion $L \subset K$ of a subcomplex is an example of a map of simplicial complexes. In particular, for any simplicial complex K, the inclusions $\mathrm{sk}_0 K \subset \mathrm{sk}_1 K \subset \cdots \subset \mathrm{sk}_k K \subset \Delta[K_0]$, are maps of simplicial complexes. Note that in this case the induced functions on the set of vertices for all these inclusions are given by the identity function $\mathrm{id}\colon K_0 \to K_0$. This is a situation in which the identity function on sets of vertices does not induce the identity function of simplicial comlexes.

Let $\phi\colon K \to L$ and $\psi\colon L \to M$ be maps between simplicial complexes. Then their composition $\psi\phi\colon K \to M$ is also a map.

For a simplicial complex K, the function $\mathrm{id}\colon K \to K$ is a map.

178. **Exercise.** Consider two partitions of (X, P) and (Y, Q) and a function $f\colon X \to Y$. Prove that there is a map $\phi\colon \Delta[P] \to \Delta[Q]$ such that $\phi_0 = f$ if and only if f preserves partitions (see Point 10).

179. **Isomorphisms.** A map $\phi\colon K \to L$ is called an **isomorphism** if there is a map $\psi\colon L \to K$ such that $\phi\psi = \mathrm{id}_L$ and $\psi\phi = \mathrm{id}_K$. Two simplicial complexes are called **isomorphic** if and only if there is an isomorphism between them.

180. **Exercise.** Consider finite subsets X and Y of $\mathcal{U}$. Show that X and Y have the same number of elements if and only if $\Delta[X]$ and $\Delta[Y]$ are isomorphic.

181. **Exercise.** Let X be a finite subset of $\mathcal{U}$ and P and Q be partitions of X. Show that (X, P) and (X, Q) are isomorphic (see Exercise 11) if and only if $\Delta[P]$ and $\Delta[Q]$ are isomorphic.

182. **Maps and components.** Let $\phi\colon K \to L$ be a map between simplicial complexes. Choose vertices v and w in K that are in the same component. We thus have a sequence $v = x_0, \ldots, x_n = w$ of vertices in K such that $\{x_i, x_{i+1}\}$ is a simplex in K. Since ϕ is a map, $\{\phi_0(x_i), \phi_0(x_{i+1})\}$ is a simplex in L. Thus by considering the sequence $\phi_0(v) = \phi_0(x_0), \ldots, \phi_0(x_n) = \phi_0(w)$ we can conclude $\phi_0(v)$ and $\phi_0(w)$ are in the same component of L. The function $\phi_0\colon K_0 \to L_0$ therefore preserves partitions $(K_0, \pi_0(K))$ and $(L_0, \pi_0(L))$, and we denote by $\pi_0(\phi)\colon \pi_0(K) \to \pi_0(L)$ the induced function (see Point 10).

183. **Homotopy relation.** Let $\psi, \phi\colon K \to L$ be maps of simplicial complexes.

- The maps ψ, ϕ are said to be **directly homotopic** if, for any simplex σ in K, the union $\psi(\sigma) \cup \phi(\sigma)$ is a simplex in L.
- The maps ψ, ϕ are said to be **homotopic** if there is a sequence of maps $\alpha_i \colon K \to L$, for $i = 0, \ldots, n$, such that $\alpha_0 = \psi$, $\alpha_n = \phi$, and α_i and α_{i+1} are directly homotopic for $i = 0, \ldots, n-1$.

Being directly homotopic is in general not an equivalence relation on maps (see Exercise 186), however being homotopic is.

184**. Exercise.**

(1) Show that being homotopic is an equivalence relation on the set of maps between two simplicial complexes.
(2) Show that if $\phi, \psi \colon K \to L$ are (directly) homotopic maps of simplicial complexes, then so are the compositions $\alpha\phi, \alpha\psi \colon K \to M$ and $\phi\beta, \psi\beta \colon M \to L$ for any maps of simplicial complexes $\alpha \colon L \to M$ and $\beta \colon M \to K$.

185**. Components and the homotopy relation.** Choose an element x in $\mathcal{U}$. Let K be a simplicial complex. For any vertex v in K, define $v \colon \Delta[x] \to K$ to be the map induced by the function $\Delta[x]_0 \ni x \mapsto v \in K_0$. This defines a bijection between the set of vertices K_0 of K and the set of maps $\Delta[x] \to K$.

Vertices typically represent points of a data sets. Therefore the relation of being in the same connected component for vertices might suggest that the data points represented by such vertices are clustered and regarded as similar. This is the reason why connected components and ability of counting them is important for us.

What is the property for maps $\Delta[x] \to K$ that corresponds to the property of vertices of being in the same connected component?

186**. Exercise.** Let v and w be vertices in a simplicial complex K and $v, w \colon \Delta[x] \to K$ be the corresponding maps.

(1) Show $v, w \colon \Delta[x] \to K$ are directly homotopic if and only if $\{v, w\}$ is a simplex in K.
(2) Prove v and w are in the same connected component of K if and only if the maps $v, w \colon \Delta[x] \to K$ are homotopic.

Vietoris-Rips complexes

187. **Vietoris-Rips complexes.** In Point 173 we presented how to think about a partition, or equivalently a discrete pseudometric on X (see Point 38), as a simplicial complex on $X \subset \mathcal{U}$. In this section we are going to explain how an arbitrary distance on X can also lead to a simplicial complex on X extending the Δ construction from Point 173. We are going to define in fact a whole family of such extensions indexed by t in $[0, \infty)$ which are called Vietoris-Rips complexes. These are reminiscent of the partitions X/d_t constructed in Point 25. If d is not an ultra pseudometric, then these partitions forget information about the distance d. Vietoris-Rips complexes on the other hand do not (see Proposition 194). All the information about the distance is contained by the Vietoris-Rips complexes. Vietoris-Rips complexes do not forget or add information about d. The advantage of transforming information about a distance into a simplicial complex is that we can then use its geometry to extract invariants. In this course we focus on invariants which are based on homology.

Let d be a distance on $X \subset \mathcal{U}$. For t in $[0, \infty)$ define $\mathrm{VR}_t(d)$ to be the collection of these finite non-empty susbsets $\{x_0, \ldots, x_n\} \subset X$ such that $d(x_i, x_j) \leq t$ for any i and j. This collection is a simplicial complex called the **Vietoris-Rips complex** of (X, d) at scale t.

188. **Exercise.** Prove that indeed $\mathrm{VR}_t(d)$ is a simplicial complex.

189. **Exercise.** A simplicial complex K is called **flag** if a set $\sigma \subset \mathcal{U}$ is a simplex in K if and only if any 2 element subset of σ is a simplex in K. Show that $\mathrm{VR}_t(d)$ is a flag. Conclude that $\mathrm{VR}_t(d) = \mathrm{VR}_t(d')$ if and only if theirs 1-st skeletons are equal $\mathrm{sk}_1\mathrm{VR}_t(d) = \mathrm{sk}_1\mathrm{VR}_t(d')$ (see Point 172).

Here is the relation between Vietoris-Rips complexes and the partitions X/d_t constructed in Point 25:

190. **Proposition.** *For any distance d on $X \subset \mathcal{U}$, the partition $\pi_0(\mathrm{VR}_t(d))$ of X (see Point 174) coincides with the partition X/d_t defined in Point 25.*

Proof. By definition (see point 174) two vertices x and y in $\mathrm{VR}_t(d)$ belong to the same connected component of $\mathrm{VR}_t(d)$ if there is a sequence of elements $x = x_0, \ldots, x_n = y$ in X such that $\{x_i, x_{i+1}\}$ is an edge in $\mathrm{VR}_t(d)$, i.e. if $d(x_i, x_{i+1}) \leq t$, for $i = 0, \ldots, k-1$. That is exactly the definition of the equivalence relation defining the partition X/d_t. $\square$

191. **Exercise.** Let d be a distance on $X \subset \mathcal{U}$. Consider the partition X/d of X.

(1) Show that $\mathrm{VR}_0(d) \subset \Delta[X/d_0] \subset \Delta[X]$.
(2) Show that if d is a pseudometric, then $\mathrm{VR}_0(d) = \Delta[X/d_0]$.

(3) Show that if X is finite, than there is a real number T such that $\mathrm{VR}_t(d) = \Delta[X]$ for all $t \geq T$.

192. Let $s \leq t$ be in $[0, \infty)$ and d be a distance on $X \subset \mathcal{U}$. By definition a subset $\{x_0, \ldots, x_n\} \subset X$ is a simplex in $\mathrm{VR}_s(d)$ if $d(x_i, x_j) \leq s$ for any i and j. Since $s \leq t$, such a subset is also a simplex in $\mathrm{VR}_t(d)$. This means that $\mathrm{VR}_s(d)$ is a sub-complex of $\mathrm{VR}_t(d)$. We call the inclusion $\mathrm{VR}_s(d) \subset \mathrm{VR}_t(d)$ the **transition map** from s to t. Thus for $0 \leq s \leq t$, we have a sequnce of transition maps:

$$\mathrm{VR}_0(d) \subset \mathrm{VR}_s(d) \subset \mathrm{VR}_t(d) \subset \Delta[X]$$

193. **Exercise.** Assume $X \subset \mathcal{U}$ is finite. Let d be a distance on X. Show that the Vietoris-Rips complexes $\mathrm{VR}_t(d)$ for t in $[0, \infty)$ satisfy the following tameness properties: there is a sequence $0 = a_0 < a_1 < \cdots < a_n$ in $[0, \infty)$ such that a transition map $\mathrm{VR}_s(d) \subset \mathrm{VR}_t(d)$ may fail to be the equality only if there is $i = 0, \ldots, n$ such that $s < a_i \leq t$.

Vietoris-Rips complexes do not forget or add information about the distance d:

194. **Proposition.** *Let d and d' be distances on $X \subset \mathcal{U}$. Then $d = d'$ if and only if $\mathrm{VR}_t(d) = \mathrm{VR}_t(d')$ for all t in $[0, \infty)$.*

Proof. If $d = d'$ then clearly $\mathrm{VR}_t(d) = \mathrm{VR}_t(d')$ for all t. We need to show the reverse implication. Assume $d \neq d'$. Let x and y be elements in X such that $d(x, y) \neq d'(x, y)$. Let $d(x, y) > d'(x, y)$. Then, for $t = d'(x, y)$, the subset $\{x, y\} \subset X$ is an edge in $\mathrm{VR}_t(d')$ but not in $\mathrm{VR}_t(d)$ and hence these simplicial complexes are different. □

195. **Proposition.** *Consider two distance spaces (X, d_X) and (Y, d_Y) with $X, Y \subset \mathcal{U}$. A function $f\colon X \to Y$ induces a map between $\mathrm{VR}_t(d_X)$ and $\mathrm{VR}_t(d_Y)$ for all t if and only if f is Lipschitz (see Point 71), i.e., if $d_X(x, y) \geq d_Y(f(x), f(y))$ for all x and y in X.*

We are going to use the symbol $\mathrm{VR}_t(f)\colon \mathrm{VR}_t(d_X) \to \mathrm{VR}_t(d_Y)$ to denote the map induced by a Lipschitz function $f\colon X \to Y$.

Proof. Assume f is Lipschitz. Consider a simplex $\sigma = \{x_0, \ldots, x_n\}$ in $\mathrm{VR}_t(d_X)$. Thus $d_X(x_i, x_j) \leq t$ for any i and j. Since f is Lipschitz, $d_Y(f(x_i), f(x_j)) \leq t$ and therefore $f(\sigma) = \{f(x_0), \ldots, f(x_n)\}$ is also a simplex in $\mathrm{VR}_t(d_Y)$. We can conclude that f indeed induces a map.

Assume $f\colon X \to Y$ induces a map between $\mathrm{VR}_t(d_X)$ and $\mathrm{VR}_t(d_Y)$ for any t. This means that for $t = d_X(x_0, x_1)$, since $\{x_0, x_1\}$ is a simplex in $\mathrm{VR}_t(d_X)$, then $\{f(x_0), f(x_1)\}$ is a simplex in $\mathrm{VR}_t(d_Y)$. This gives $d_Y(f(x_0), f(x_1)) \leq t = d_X(x_0, x_1)$ and hence f is Lipschitz. □

196. **Theorem.** *Let (X, d_X) and (Y, d_Y) be distance spaces with $X, Y \subset \mathcal{U}$. A function $f\colon X \to Y$ is an isometry (see Point 71) if and only if the map $\mathrm{VR}_t(f)\colon \mathrm{VR}_t(d_X) \to \mathrm{VR}_t(d_Y)$ is an isomorphism for all t in $[0, \infty)$.*

Proof. If $f\colon X \to Y$ is a bijection that preserves distances, then its inverse $f^{-1}\colon Y \to X$ also preserves distances. According to Proposition 195, the functions f and f^{-1} induce maps $\mathrm{VR}_t(f)\colon \mathrm{VR}_t(d_X) \to \mathrm{VR}_t(d_Y)$ and $\mathrm{VR}_t(f^{-1})\colon \mathrm{VR}_t(d_Y) \to \mathrm{VR}_t(d_X)$ for any t. Since these maps are inverse to each other $\mathrm{VR}_t(f)$ is an isomorphism.

Assume $\mathrm{VR}_t(f)\colon \mathrm{VR}_t(d_X) \to \mathrm{VR}_t(d_Y)$ is an isomorphism for any t. Then f is a bijection since it has an inverse function. We can now use Proposition 195 to get that f and f^{-1} are Lipschitz. This implies that f preserves the distances. □

Vietoris-Rips construction converts a proximity relation between Lipschitz functions into homotopy relation. That is the key property why this construction is so useful from the topology point of view. Imprecision and noise is translated into homotopy relation.

197. **Theorem.** *Let (X, d_X) and (Y, d_Y) be distance spaces and let $f, g\colon X \to Y$ be Lipschitz functions. Assume:*

- *Y is a pseudometric;*
- *$d_Y(f(x), g(x)) \leq \epsilon$ for all x in X.*

Then, for all t in $[0, \infty)$, the following two compositions from $\mathrm{VR}_t(X)$ into $\mathrm{VR}_{t+\epsilon}(Y)$ are directly homotopic (see Point 183):

$$\mathrm{VR}_t(X) \underset{\mathrm{VR}_t(g)}{\overset{\mathrm{VR}_t(f)}{\rightrightarrows}} \mathrm{VR}_t(Y) \hookrightarrow \mathrm{VR}_{t+\epsilon}(Y)$$

Proof. Let $\sigma = \{x_0, \ldots, x_n\}$ be a simplex in $\mathrm{VR}_t(X)$. By definition it means that $d_X(x_i, x_j) \leq t$ for all i and j. We need to show that the following set is a simplex in $\mathrm{VR}_{t+\epsilon}(Y)$:

$$f(\sigma) \cup g(\sigma) = \{f(x_0), \ldots, f(x_n)\} \cup \{g(x_0), \ldots, g(x_n)\}$$

Since f and g are Lipschitz:

$$d_Y(f(x_i), f(x_j)) \leq d_X(x_i, x_j) \leq t \leq t + \epsilon$$
$$d_Y(g(x_i), g(x_j)) \leq d_X(x_i, x_j) \leq t \leq t + \epsilon$$

Furthermore, we can use the assumption (the triangle inequality on Y and proximity between f and g) to also get:

$$d_Y(f(x_i), g(x_j)) \leq d_Y(f(x_i), f(x_j)) + d_Y(f(x_j), g(x_j)) \leq t + \epsilon$$

□

Homology

198. Fix a field K. Let X be a simplicial complex. Recall that X_n denotes the set of n-dimensional simplices of X. For example X_0 is the set of vertices and X_1 the set of edges (simplices of dimension 1). Assume $X_0 = \{v_0, \ldots, v_k\}$. Here is a step by step process of how to define homology $H_n(X, K)$ of X with coefficients in K for any integer $n \geq 0$:

Step 1. Choose an ordering $v_0 < v_1 < \cdots < v_k$ of the set X_0 of vertices of X. If such an ordering is chosen, then the set X_n of n-dimensional simplices in X can be identified with the set of increasing sequences $\sigma = (x_0 < x_1 < \cdots < x_n)$ of $n+1$ elements in X_0.

Step 2. We use the above choice of ordering to define a function:

$$d_i \colon X_n \to X_{n-1}$$

for any $n \geq 1$ and $0 \leq i \leq n$. This function maps an n dimensional simplex $\sigma = (x_0 < x_1 < \cdots < x_n)$ to an $n-1$ dimensinal simplex which is formed by removing σ's i-th vertex (the hat over x_i in the expression below means remove this vertex):

$$d_i\sigma := (x_0 < \cdots < \widehat{x_i} < \cdots < x_n)$$

Being able to say which vertex in a simplex is the i-th vertex is the reason why an ordering on the vertices needs to be chosen. Note that if $0 \leq j < i \leq n$, then:

$$d_j d_i \sigma = d_{i-1} d_j \sigma$$

Step 3. For every n, consider the vector space KX_n associated with the set X_n. Consider next the linear functions $Kd_i \colon KX_n \to KX_{n-1}$ for any $n \geq 1$ and $0 \leq i \leq n$. Define a linear function $\delta_n \colon KX_n \to KX_{n-1}$ to be the alternating sum:

$$\delta_n = Kd_0 - Kd_1 + Kd_2 - \cdots (-1)^i Kd_i \cdots (-1)^n Kd_n$$

Furthermore set $\delta_0 \colon KX_0 \to 0$ to be the zero function. For example, if v and w are vertices in X, then $\delta_1(v < w) = w - v$.

We have defined a sequence of linear functions:

$$\cdots \xrightarrow{\delta_{n+1}} KX_n \xrightarrow{\delta_n} KX_{n-1} \xrightarrow{\delta_{n-1}} \cdots \xrightarrow{\delta_2} KX_1 \xrightarrow{\delta_1} KX_0 \xrightarrow{\delta_0 = 0} 0$$

Note that in the case the considered field contains the field $\mathbf{F}_2$ (which is equivalent to $2 = 0$ in K), then δ_n does not depend on the choice of the ordering in Step 1. Thus if we consider only coefficients K containing $\mathbf{F}_2$ to define homology we can avoid steps one and two and define directly:

$$\delta_n(\sigma) = \sum_{x \in \sigma} \sigma \setminus \{x\}$$

The linear function $\delta_n\colon KX_n \to KX_{n-1}$ is called the **boundary** function. Elements of $\ker(\delta_n)$ are called n**-cycles** or simply **cycles** if the dimension is assumed. Elements of $\mathrm{im}(\delta_{n+1})$ are called n**-boundaries** or simply **boundaries** in case the dimension is assumed. It turns out that any boundary element is a cycle:

199**. Proposition.** *The composition $\delta_n\delta_{n+1}$ is the zero function.*

Proof. Consider the element $\sigma = (x_0 < x_1 < \cdots < x_{n+1})$ in KX_{n+1}. By definition:

$$\delta_{n+1}(\sigma) = \sum_{i=0}^{n+1}(-1)^i d_i\sigma$$

Thus:

$$\delta_n\delta_{n+1}(\sigma) = \sum_{j=0}^{n}(-1)^j\sum_{i=0}^{n+1}(-1)^i d_j d_i\sigma = \sum_{0\le j<i\le n+1}(-1)^{i+j}d_jd_i\sigma + \sum_{0\le i\le j<n+1}(-1)^{i+j}d_jd_i\sigma =$$
$$= \sum_{0\le j<i\le n+1}(-1)^{i+j}d_{i-1}d_j\sigma + \sum_{0\le i\le j<n+1}(-1)^{i+j}d_jd_i\sigma =$$
$$= \sum_{0\le j\le i\le n}(-1)^{i+j+1}d_id_j\sigma + \sum_{0\le i\le j<n+1}(-1)^{i+j}d_jd_i\sigma = 0$$

□

According to the proposition $\mathrm{im}(\delta_{n+1}) \subset \ker(\delta_n)$, i.e., any boundary element is a cycle. The reverse implication is not true in general. Not every cycle is a boundary. Homology measures how many more cycles than boundary elements are there:

200**. Definition.** The quotient of $\ker(\delta_n)$ by $\mathrm{im}(\delta_{n+1})$ is called the n**-th homology of** X:

$$H_n(X,K) := \ker(\delta_n)/\mathrm{im}(\delta_{n+1})$$

According to this definition $H_n(X,K) = 0$ if and only if any n-th cycle is a boundary element.

201**. Exercise.** Show that $H_n(\Delta[0],K) = \begin{cases} K & \text{if } n = 0 \\ 0 & \text{if } n > 0 \end{cases}$.

202**. Exercise.** Let X and Y be disjoint simplicial complexes. Show that $H_n(X \coprod Y, K)$ and $H_n(X,K) \oplus H_n(Y,K)$ are isomorphic.

203**.** We think about homology as a simplifying tool. It translates simplicial complexes, which are complicated objects, into more familiar and much easier to understand and study vector spaces.If X is a finite simplicial complex, then $H_n(X,F)$ is finite dimensional for any n. We can thus think about homology as a function:

$$\boxed{\text{Finite simplicial complexes}} \xrightarrow{H_n(-,F)} \boxed{\text{finite dimensional vector spaces}}$$

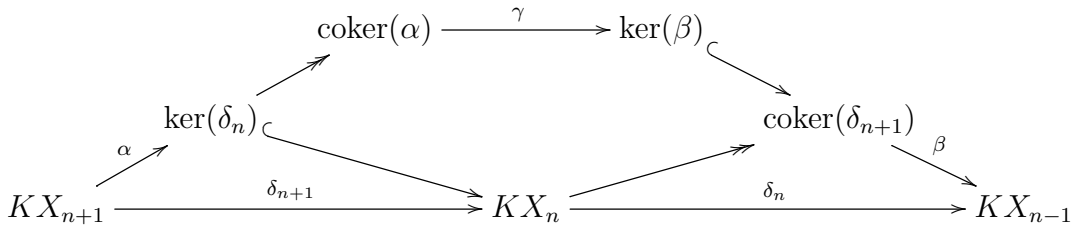
Here are some fundamental questions that need to be adressed:

- What role the choice of the orderings of the finite sets play?
- What information homology encodes and forgets? We already know that homology measures how many more cycles than boundaries are there, but what does this mean?
- What information can we extract from the fact that $H_n(X, K)$ is of a given dimension?

204**. Exercise.** Let X be a simplicial complex. Choose an ordering of its set of vertices X_0 and consider the linear maps:

$$KX_{n+1} \xrightarrow{\delta_{n+1}} KX_n \xrightarrow{\delta_n} KX_{n-1}$$

We can extend this diagram by taking kernels and cokernels to a commutative diagram where all the additional functions α,β and γ are uniquely determined by the commutativity requirement:



Prove that γ is an isomorphism and that both $\operatorname{coker}(\alpha)$ and $\ker(\beta)$ are isomorphic to $H_n(X, K)$.

205**. Exercise.** Consider the following matrix:

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 3 \\ 2 & 2 & 3 & 0 \end{bmatrix}$$

(1) Show that this matrix defines a pseudo metric d on a set with 4 elements.
(2) Describe $\mathrm{VR}_0(d)$, $\mathrm{VR}_1(d)$, $\mathrm{VR}_2(d)$, and $\mathrm{VR}_3(d)$.
(3) Calculate $H_n(\mathrm{VR}_t(d), \mathbf{F}_2)$ for $n = 0, 1$ and $t = 0, 1, 2, 3$.

206**.** Let X be a simplicial complex. Choose an ordering of its set of vertices X_0. Recall that $H_0(X, K)$ is the cokernel of the function $\delta_1 \colon KX_1 \to KX_0$. Since vertices form a basis in KX_0, their equivalence classes in $H_0(X, K)$ generate this vector spaces. Thus $H_0(X, K)$ is generated by homology classes of the form $[v]$ where v is a vertex of X.

Consider the inclusion $X_0 \subset KX_0$ of the basis given by the vertices. Recall that two vertices v and w of X lie in the same connected component if and only if there is a sequence $v = x_0, \ldots, x_n = w$ of vertices in X such that $\{x_i, x_{i+1}\}$ is a simplex in X for

any i. For any such sequence define:

$$c_i = \begin{cases} 1 & \text{if } x_i < x_{i+1} \\ -1 & \text{if } x_i > x_{i+1} \end{cases}$$

Thus if v and w are in the same component, then:

$$\delta_1 \left(c_0\{x_0, x_1\} + \cdots + c_{n-1}\{x_{n-1}, x_n\} \right) = x_1 - x_0 + x_2 - x_1 + \cdots + x_n + x_{n-1} = w - v$$

and hence $v - w$ is a boundary in KX_0 and consequently the homology classes $[v]$ and $[w]$ in $H_0(X, K)$ are equal. In this way we get a commutative diagram:

$$\begin{array}{ccc} & & KX_1 \\ & & \downarrow{\scriptstyle \delta_1} \\ X_0 & \hookrightarrow & KX_0 \\ \downarrow & & \downarrow \\ \pi_0(X) & \longrightarrow & H_0(X, K) \end{array}$$

We claim that the constructed function $\pi_0(X) \to H_0(X, K)$ is an inclusion:

207. **Proposition.** *Two vertices v and w of X lie in the same connected component of X if and only if the homology clases $[v]$ and $[w]$ in $H_0(X, K)$ are equal.*

Proof. Assume $[v] = [w]$. If $v = w$, there is nothing to prove. Assume $v \neq w$.

The assumption $[v] = [w]$ implies there is a sequence of edges $\{s_1 < e_1\}, \ldots, \{s_n < e_n\}$ in X such that:

$$\delta_1(\{s_1 < e_1\} + \cdots + \{s_n < e_n\} = e_1 - s_1 + \cdots + e_n - s_n = e_1 + \cdots + e_n - (s_1 + \cdots + s_n) = v - w$$

As a consequewnce we obtained the following set equality

$$\{e_1, \ldots, e_n, w\} = \{s_1, \ldots, s_n, v\}$$

By permuting if necessary, we could therefore assume: $e_n = v$, $e_{n-1} = s_n$, $\ldots$, $e_1 = s_2$ and $w = s_1$. This implies v and w belong to the same connected component of X. $\square$

208. **Exercise.** Prove that $H_0(X, K)$ and $K\pi_0(X)$ are isomorphic vector spaces.

Maps and their homology.

209. Let X and Y be simplicial complexes. Choose orderings of their sets of vertices X_0 and Y_0. This is the first step needed to define the homologies of X and Y (see 198).

Let $f\colon X \to Y$ be a map. Our aim is to explain how such a map leads to a linear function between homologies of X and Y which we denoted by the symbol $H_n(f,K)\colon H_n(X,K) \to H_n(Y,K)$. Here are all the steps needed for this definition:

210. **Step 1.** Let $n \geq 0$. Choose a simplex $\sigma = (v_0 < v_1 < \cdots < v_n)$ in X_n. We are going to define an element $c(f,\sigma)$ in K which is either 0, or 1 or -1. Consider the simplex $f(\sigma) = \{f(v_0), \ldots, f(v_n)\}$ in Y. There are two possibilities:

Case 1. Assume $|f(\sigma)| < n+1$. In this case f identifies some of the vertices of σ or equivalently $\dim(f(\sigma)) < \dim(\sigma)$. We define:

$$c(f,\sigma) := 0$$

Case 2. Assume $|f(\sigma)| = n+1$. In this case all vertices in σ are mapped via f to different vertices in Y or equivalently $\dim(f(\sigma)) = \dim(\sigma)$. Let α be the unique permutation of the set $\{0,\ldots,n\}$ for which the following order relations with respect to chosen ordering on Y_0 hold:

$$f(v_{\alpha(0)}) < f(v_{\alpha(1)}) < \cdots < f(v_{\alpha(n)})$$

Define $c(f,\sigma)$ to be the signature of the permutation α. Thus:

$$c(f,\sigma) := \begin{cases} 1 & \text{if } \alpha \text{ can be expressed as a product of even number of transpositions} \\ -1 & \text{if } \alpha \text{ can be expressed as a product of odd number of transpositions} \end{cases}$$

Observe that if $n = 0$, then $c(f,v) = 1$ for any vertex v in X.

Note also if the chosen field contains $\mathbf{F}_2$, then since $1 = -1$:

$$c(f,\sigma) = \begin{cases} 0 & \text{if } \dim(f(\sigma)) < \dim(\sigma) \\ 1 & \text{if } \dim(f(\sigma)) = \dim(\sigma) \end{cases}$$

211. **Step 2.** Define $f_n\colon KX_n \to KY_n$ to be the unique linear function such that for a simplex σ in X_n:

$$f_n(\sigma) := \begin{cases} 0 & \text{if } c(f,\sigma) = 0 \text{ i.e., if } \dim(f(\sigma)) < \dim(\sigma) \\ c(f,\sigma)f(\sigma) & \text{otherwise} \end{cases}$$

In principle f_n maps a simplex σ to the simplex $f(\sigma)$ modified by a coefficient that is either 0, or 1 or -1. For example if the chosen field contains the field with two

elements $\mathbf{F}_2$, then since $1 = -1$:

$$f_n(\sigma) := \begin{cases} 0 & \text{if } \dim(f(\sigma)) < \dim(\sigma) \\ f(\sigma) & \text{otherwise} \end{cases}$$

212. **Step 3.** *The functions f_n make the following diagram commutative:*

$$\begin{array}{ccccccccccccc} \cdots & \xrightarrow{\delta_{n+1}} & KX_n & \xrightarrow{\delta_n} & KX_{n-1} & \xrightarrow{\delta_{n-1}} & \cdots & \xrightarrow{\delta_2} & KX_1 & \xrightarrow{\delta_1} & KX_0 & \xrightarrow{\delta_0=0} & 0 \\ & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \\ \cdots & \xrightarrow{\delta_{n+1}} & KY_n & \xrightarrow{\delta_n} & KY_{n-1} & \xrightarrow{\delta_{n-1}} & \cdots & \xrightarrow{\delta_2} & KY_1 & \xrightarrow{\delta_1} & KY_0 & \xrightarrow{\delta_0=0} & 0 \end{array}$$

Proof. We need to show that for any simplex σ in X_n:

$$\delta_n f_n(\sigma) = f_{n-1}\delta_n(\sigma)$$

Let $\sigma = (v_0 < \cdots < v_n)$. We consider three cases:

Case 1. Assume $|f(\sigma)| \leq n-1$. In this case $f_n(\sigma) = 0$. For any i, the simplex $f(d_i\sigma)$ is of dimension non exceeding the dimension of $f(\sigma)$ which according to our assumption is at most $n-2$. Thus in this case also $f_{n-1}(d_i\sigma) = 0$. We can conclude $f_{n-1}(\delta_n(\sigma)) = 0$. Thus both of the compositions $f_{n-1}\delta_n$ and $\delta_n f_n$ map σ to 0.

Case 2. Assume $|f(\sigma)| = n$. In this case $f_n(\sigma) = 0$. Thus we need to show that also $f_{n-1}\delta_n(\sigma) = 0$. The assumption implies that only two vertices of σ, say $v_a < v_b$, are mapped via f to the same vertex of Y. It follows that if $i \neq a$ and $i \neq b$, then $f_{n-1}(d_i\sigma) = 0$. Thus:

$$f_{n-1}(\delta_n(\sigma)) = (-1)^a f_{n-1}(d_a\sigma) + (-1)^b f_{n-1}(d_b\sigma) =$$

$$= (-1)^a c(f, d_a\sigma) f(d_a\sigma) + (-1)^b c\left(f, d_b\sigma\right) f(d_b\sigma)$$

Since $f(v_a) = f(v_b)$, then $f(d_a\sigma) = f(d_b\sigma)$ and hence:

$$f_{n-1}(\delta_n(\sigma)) = \left((-1)^a c(f, d_a\sigma) + (-1)^b c(f, d_b\sigma)\right) f(d_a\sigma)$$

We claim that:

$$c(f, d_b\sigma) = (-1)^{b-a-1} c(f, d_a\sigma)$$

This would then imply:

$$(-1)^a c(f, d_a\sigma) + (-1)^b c(f, d_b\sigma) = 0$$

proving:

$$f_{n-1}(\delta_n(\sigma)) = 0$$

To see the claim note that $c(f, d_b\sigma)$ is the signature of the permutation that orders:

$$(f(v_0), \ldots, f(v_a) \ldots, \widehat{f(v_b)}, \ldots f(v_n))$$

and $c\,(f, d_a(\sigma))$ is the signature of the permutation that orders:

$$(f(v_0), \dots, \widehat{f(v_a)}, \dots, f(v_b), \dots f(v_n))$$

One of these permutations can be obtained from the other by multiplying it with the permutation that moves $f(v_b)$ through the elements $f(v_{a+1}), \dots, f(v_{b-1})$. Since this last permutation has signature $(-1)^{b-a-1}$, the claim follows.

Case 3. Assume $|f(\sigma)| = n+1$. Let α be the permutation that orders the sequence:

$$\big(f(v_{\alpha(0)}) < \dots < f(v_{\alpha(n)})\big)$$

It is then clear that:

$$d_i f(\sigma) = f(d_{\alpha(i)}\sigma)$$

Set $c = c(f, \sigma)$ to the the signature of α. For any i the signature of the permutation γ that orders vertices of $f(d_{\alpha(i)})$ is equal to $c(-1)^{i-\alpha(i)}$. This is because such a permutation has the same signature as the permutation of $0, \dots, n$ that maps $\alpha(i)$ to itself and on the rest coincides with γ. This extended permutation is a product of α and the permutation that moves $\alpha(i)$ through all elements between i and $\alpha(i)$. Its signature is therefore equal to $c(-1)^{\alpha(i)-i}$, as claimed. We can now conclude:

$$\delta_n f_n(\sigma) = \Sigma_{i=0}^n (-1)^i d_i f_n(\sigma) = \Sigma_{i=0}^n (-1)^i d_i c f(\sigma) = \Sigma_{i=0}^n (-1)^i c f(d_{\alpha(i)}\sigma) =$$

$$= \Sigma_{i=0}^n (-1)^i (-1)^{\alpha(i)-i} c (-1)^{\alpha(i)-i} f(d_{\alpha(i)}\sigma) = \Sigma_{i=0}^n (-1)^{\alpha(i)} f_{n-1}(d_{\alpha(i)}\sigma) =$$

$$= f_{n-1}\left(\Sigma_{i=0}^n (-1)^{\alpha(i)} d_{\alpha(i)}\sigma\right) = f_{n-1}(\delta_n(\sigma))$$

□

213. **Corollary.** *$f_n \colon KX_n \to KY_n$ maps cycles to cycles and boundaries to boundaries.*

Proof. Let x in KX_n be a cycle. This means that $\delta_n(x) = 0$. We can then use Proposition 212 to get $\delta_n(f_n(x)) = f_{n-1}(\delta_n(x)) = f_{n-1}(0) = 0$ and conclude that $f_n(x)$ is also a cycle.

Assume x is a boundary, which means that $x = \delta_{n+1}(y)$ for some y in KX_{n+1}. Proposition 212 implies that $\delta_{n+1}(f_{n+1}(y)) = f_n(\delta_{n+1}(y)) = f_n(x)$ and we can conclude $f_n(x)$ is also a boundary. □

214. **Step 4.** According to Corollary 213:

(1) If x in KX_n is a cycle, then so is $f_n(x)$ in KY_n.
(2) If x is a boundary in KX_n, then so is $f_n(x)$ in KY_n.

These are the key properties needed to define the function $H_n(f, K) \colon H_n(X, K) \to H_n(Y, K)$. Recall:

$$H_n(X, K) = \ker(\delta_n \colon KX_n \to KX_{n-1}) / \mathrm{im}(\delta_{n+1} \colon KX_{n+1} \to KX_n)$$

$$H_n(Y, K) = \ker(\delta_n \colon KY_n \to KY_{n-1}) / \mathrm{im}(\delta_{n+1} \colon KY_{n+1} \to KY_n)$$

Let $[x] \in H_n(X, K)$ be the homology class defined by a cycle x in KX_n. Since f_n preserves cycles, $f_n(x)$ is a cycle in KY_n. We can therefore consider its homology class $[f_n(x)]$ in $H_n(Y, K)$ and define:

$$H_n(f, K)\colon H_n(X, K) \to H_n(Y, K)$$

to be the function that maps the homology class $[x]$ to the homology class $[f_n(x)]$. Since f_n preserves boundaries, the association $[x] \mapsto [f_n(x)]$ does not depend on the choice of the representative x of the homology class $[x]$ and hence the function $H_n(f, K)$ is well defined.

It is important for us to understand how homology behaves with respect to compositions of maps of simplicial complexes:

215. **Proposition.** *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps of simplicial complexes. Then $H_n(gf, K) = H_n(g, K)H_n(f, K)$,i.e., the following diagram commutes:*

$$H_n(X, K) \xrightarrow{H_n(f,K)} H_n(Y, K) \xrightarrow{H_n(g,K)} H_n(Z, K)$$
$$\qquad\qquad \underset{H_n(gf,K)}{\searrow\!\!\nearrow}$$

Proof. To show the proposition we need to prove $g_n f_n = (gf)_n$. This follows directly from: □

216. **Lemma.** *For any simplex σ in X, $c(gf, \sigma) = c\,(g, f(\sigma))\, c(f, \sigma)$*

Proof. If $\dim(f(\sigma)) < \dim(\sigma)$ or $\dim(gf(\sigma)) < \dim(f(\sigma))$, then $\dim(gf(\sigma)) < \dim(\sigma)$. Thus if $c(f, \sigma) = 0$ or $c\,(g, f(\sigma)) = 0$, then $c(gf, \sigma) = 0$.

Assume $\dim(f(\sigma)) = \dim(\sigma)$ and $\dim(gf(\sigma)) = \dim(f(\sigma))$. Let $\sigma = (v_0 < \cdots < v_n)$. The permutation needed to order the elements $(gf(v_0), \ldots, gf(v_n))$ can be described as a product of the following two permutations. The first is the permutation α for which $(f(v_{\alpha(0)}) < \cdots < f(v_{\alpha(0)}))$. The second permutation orders $(gf(v_{\alpha(0)}), \ldots, f(v_{\alpha(0)}))$. As the signature of a product of permutations is a product of their signatures we obtain the desired equality $c(gf, \sigma) = c\,(g, f(\sigma))\, c(f, \sigma)$. □

Another fact we are going to use and which follows directly from the definition is:

217. **Proposition.** *Let X be a simplicial complex. For a fixed ordering on X_0, $H_n(\mathrm{id}, K)\colon H_n(X, K) \to H_n(X, K)$ is the identity function.*

Here are some fundamental consequences of Propositions 215 and 217:

218. **Corollary.**

(1) *If $f\colon X \to Y$ is an isomorphism of simplicial complexes, then the function $H_n(f, K)\colon H_n(X, K) \to H_n(Y, K)$ is an isomorphism, for all choices of orderings on X_0 and Y_0.*

(2) *Let X be a simplicial complex. For any two choices of orderings on X_0, the function $H_n(\mathrm{id}, K)\colon H_n(X, K) \to H_n(X, K)$ is an isomorphism.*
(3) *If X and Y are isomorphic simplicial complexes, then so are the vector spaces $H_n(X, K)$ and $H_n(Y, K)$.*

Homotopy and homology.

The fundamental property of homology is:

219. **Theorem.** *Let $f, g\colon X \to Y$ be maps between simplicial complexes. If f, g are directly homotopic, then $H_n(f, K) = H_n(g, K)$.*

220. Let X be a subset in $\mathcal{U}$ and x be an element in X. Consider the map $x\colon \Delta[0] \to \Delta[X]$. The following composition is the identity:

$$\Delta[0] \xrightarrow{v} \Delta[X] \longrightarrow \Delta[0]$$

The other composition:

$$\Delta[X] \longrightarrow \Delta[0] \xrightarrow{v} \Delta[X]$$

however is not the identity, unless $k = 0$. This last composition although different from $\mathrm{id}\colon \Delta[X] \to \Delta[X]$, is directly homotopic to this identity map. We can therefore conclude that on homology we get that the function $H_n(v, K)\colon H_n(\Delta[0], K) \to H_n(\Delta[X], K)$ is an isomorphism for any n. Since $H_n(\Delta[0], K) = 0$ for $n > 0$ and $H_0(\Delta[0], K) = K$, we can conclude:

$$H_n(\Delta[X], F) = \begin{cases} K & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

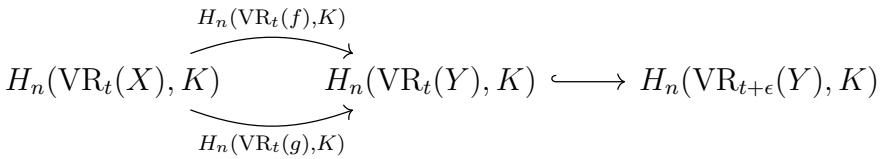
221. Let $P = \{B_1, \ldots, B_k\}$ be a partition of X. Consider the simplicial complex $\Delta[P]$ which is the disjoint union of standard simplices $\coprod_{i=1}^{k} \Delta[B_i]$. By the same argument as above one can show that the map $\coprod_{i=1}^{k} \Delta[B_i] \to \coprod_{i=1}^{k} \Delta[0]$ induces an isomorphism on homology. We can then conclude that (compare with Exercise 202):

$$H_n(\Delta[P], K) = \begin{cases} K^k & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

222. Let (X, d_X) be a distance space and (Y, d_Y) a pseudometric space. Let $f, g\colon X \to Y$ be Lipschitz functions such that $d_Y(f(x), g(x)) \leq \epsilon$ for all x in X. Then according to Theorem 197, for all t in $[0, \infty)$, the following two compositions from $\mathrm{VR}_t(X)$ into $\mathrm{VR}_{t+\epsilon}(Y)$ are directly homotopic

$$\mathrm{VR}_t(X) \underset{\mathrm{VR}_t(g)}{\overset{\mathrm{VR}_t(f)}{\rightrightarrows}} \mathrm{VR}_t(Y) \hookrightarrow \mathrm{VR}_{t+\epsilon}(Y)$$

We can then conclude that the induced maps on homology are equal:


$H_n(\mathrm{VR}_t(f),K)$
$H_n(\mathrm{VR}_t(X),K)$
$H_n(\mathrm{VR}_t(Y),K)$
$\hookrightarrow$
$H_n(\mathrm{VR}_{t+\epsilon}(Y),K)$
$H_n(\mathrm{VR}_t(g),K)$